Proofs of Statements

EC.1. Causal Transport Distance

LEMMA EC.1 **(Equivalent Definition)**. *Let $\gamma \in \Gamma(\widehat{\mathbb{P}}, \mathbb{P})$ be a transport plan. Then the following are equivalent.*

(I) $\gamma \in \Gamma_c(\widehat{\mathbb{P}}, \mathbb{P})$.

(II) *For $\widehat{\mathbb{P}}$-almost every $(\widehat{X}, \widehat{Z}) \in \mathcal{X} \times \mathcal{Z}$,*

$$\gamma_{X|(\widehat{X},\widehat{Z})} = \gamma_{X|\widehat{X}}.$$

(III) *Let $\mathrm{Proj}_X : \mathcal{X} \times \mathcal{Z} \to \mathcal{X}$ be the projection into X coordinate. For $\widehat{\mathbb{P}}$-almost every $(\widehat{x}, \widehat{z}_1), (\widehat{x}, \widehat{z}_2) \in \mathcal{X} \times \mathcal{Z}$,*

$$(\mathrm{Proj}_X)_\# \gamma(\mathrm{d}x|\widehat{x}, \widehat{z}_1) = (\mathrm{Proj}_X)_\# \gamma(\mathrm{d}x|\widehat{x}, \widehat{z}_2).$$

(IV) *For $\widehat{\mathbb{P}}_{\widehat{X}}$-almost every $\widehat{X}$ and $\mathbb{P}_X$-almost every X,*

$$\gamma_{\widehat{Z}|(\widehat{X},X)} = \gamma_{\widehat{Z}|\widehat{X}} = \widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}.$$

(V) *Let $\mathrm{Proj}_{\widehat{Z}} : \mathcal{Z} \times \mathcal{Z} \to \mathcal{Z}$ be the projection into $\widehat{Z}$ coordinate: $\mathrm{Proj}_{\widehat{Z}}(\widehat{z}, z) = \widehat{z}$. For $\widehat{\mathbb{P}}_{\widehat{X}}$-almost every $\widehat{x} \in \mathcal{X}$ and $\mathbb{P}_X$-almost every $x_1, x_2 \in \mathcal{X}$,*

$$(\mathrm{Proj}_{\widehat{Z}})_\# \gamma(\mathrm{d}\widehat{z}|\widehat{x}, x_1) = (\mathrm{Proj}_{\widehat{Z}})_\# \gamma(\mathrm{d}\widehat{z}|\widehat{x}, x_2).$$

Moreover, $\gamma \in \Gamma(\widehat{\mathbb{P}}, \mathbb{P})$ plus any one from the above is equivalent to $\gamma \in \mathcal{P}((\mathcal{X} \times \mathcal{Z}) \times (\mathcal{X} \times \mathcal{Z}))$, satisfying

(VI) *γ has a decomposition into successive regular kernels*

$$\gamma(\mathrm{d}\widehat{x}\,\mathrm{d}\widehat{z}\,\mathrm{d}x\,\mathrm{d}z) = \gamma_1(\mathrm{d}\widehat{x}\,\mathrm{d}x)\gamma_2(\mathrm{d}\widehat{z}\,\mathrm{d}z|\widehat{x}, x)$$

satisfying

$$\begin{aligned} \gamma_1 &\in \Gamma(\widehat{\mathbb{P}}_{\widehat{X}}, \mathbb{P}_X), \\ (\mathrm{Proj}_{\widehat{Z}})_\# \gamma_2(\mathrm{d}\widehat{z}|\widehat{x}, x) &= \widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}(\mathrm{d}\widehat{z}|\widehat{x}) \qquad \textit{for } \gamma_1\textit{-almost every } (\widehat{x}, x), \\ (\mathrm{Proj}_{(X,Z)})_\# \gamma_{Z|X}(\mathrm{d}z|x) &= \mathbb{P}_{Z|X}(\mathrm{d}z|x) \qquad \textit{for } \mathbb{P}_X\textit{-almost every } x. \end{aligned}$$

That is,

$$\gamma_1 \in \Gamma(\widehat{\mathbb{P}}_{\widehat{X}}, \mathbb{P}_X), \qquad \gamma_2 \in \Gamma(\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}, \mathbb{Q}^{(\widehat{X})}) \textit{ where } \mathbb{E}_{\widehat{X}\sim(\gamma_1)_{\widehat{X}|X}}[\mathbb{Q}^{(\widehat{X})}|X] = \mathbb{P}_{Z|X}.$$

Proof. The equivalence of (I), (II), and (IV) follows from the definition. It is also easy to check from the definition that (II) is equivalent to (III), and (IV) is equivalent to (V).

Suppose (VI) holds, then projecting γ onto $(X, \widehat{X}, \widehat{Z})$ coordinate, we have

$$(\mathrm{Proj}_{(X,\widehat{X},\widehat{Z})})_{\#}\gamma(\mathrm{d}x\,\mathrm{d}\widehat{x}\,\mathrm{d}\widehat{z}) = \gamma_1(\mathrm{d}\widehat{x}\,\mathrm{d}x)\cdot(\mathrm{Proj}_{\widehat{Z}})_{\#}\gamma_2(\mathrm{d}\widehat{z}|\widehat{x},x) = \gamma_1(\mathrm{d}\widehat{x}\,\mathrm{d}x)\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}(\mathrm{d}\widehat{z}|\widehat{x}).$$

Projecting onto $(\widehat{X}, \widehat{Z})$ yields

$$(\mathrm{Proj}_{(\widehat{X},\widehat{Z})})_{\#}\gamma(\mathrm{d}\widehat{x}\,\mathrm{d}\widehat{z}) = (\mathrm{Proj}_{\widehat{X}})_{\#}\gamma_1(\mathrm{d}\widehat{x})\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}(\mathrm{d}\widehat{z}|\widehat{x}) = \widehat{\mathbb{P}}_{\widehat{X}}(\mathrm{d}\widehat{x})\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}(\mathrm{d}\widehat{z}|\widehat{x}) = \widehat{\mathbb{P}}(\widehat{x},\widehat{z}).$$

As for the other marginal,

$$(\mathrm{Proj}_{(X,Z)})_{\#}\gamma(\mathrm{d}x\,\mathrm{d}z) = (\mathrm{Proj}_{X})_{\#}\gamma_1(\mathrm{d}x)\cdot(\mathrm{Proj}_{(X,Z)})_{\#}\gamma_{Z|X}(\mathrm{d}z|x) = \mathbb{P}_X(\mathrm{d}x)\mathbb{P}_{Z|X}(\mathrm{d}z|x) = \mathbb{P}(\mathrm{d}x\,\mathrm{d}z).$$

So indeed we have $\gamma \in \Gamma(\widehat{\mathbb{P}}, \mathbb{P})$. □

Proof of Lemma 1. Since $\gamma^{(q)}$ are transport plans starting from $\widehat{\mathbb{P}}$,

$$\gamma^{(q)}_{(\widehat{X},\widehat{Z})} = \widehat{\mathbb{P}}, \qquad \gamma^{(q)}_{\widehat{X}} = \widehat{\mathbb{P}}_{\widehat{X}}, \qquad \forall q \in [0,1].$$

Together with

$$\gamma^{(q)}_{(X,\widehat{X},\widehat{Z})} = (1-q)\gamma^{(0)}_{(X,\widehat{X},\widehat{Z})} + q\gamma^{(1)}_{(X,\widehat{X},\widehat{Z})}, \qquad \gamma^{(q)}_{(X,\widehat{X})} = (1-q)\gamma^{(0)}_{(X,\widehat{X})} + q\gamma^{(1)}_{(X,\widehat{X})},$$

we know that

$$\gamma^{(q)}_{X|(\widehat{X},\widehat{Z})} = (1-q)\gamma^{(0)}_{X|(\widehat{X},\widehat{Z})} + q\gamma^{(1)}_{X|(\widehat{X},\widehat{Z})}, \qquad \gamma^{(q)}_{X|\widehat{X}} = (1-q)\gamma^{(0)}_{X|\widehat{X}} + q\gamma^{(1)}_{X|\widehat{X}}.$$

Because $\gamma^{(0)}$ and $\gamma^{(1)}$ are causal, by equivalent definition (II), for $\widehat{\mathbb{P}}$-almost every $(\widehat{X}, \widehat{Z}) \in \mathcal{X} \times \mathcal{Z}$,

$$\gamma^{(0)}_{X|(\widehat{X},\widehat{Z})} = \gamma^{(0)}_{X|\widehat{X}}, \qquad \gamma^{(1)}_{X|(\widehat{X},\widehat{Z})} = \gamma^{(1)}_{X|\widehat{X}}.$$

Therefore

$$\gamma^{(q)}_{X|(\widehat{X},\widehat{Z})} = \gamma^{(q)}_{X|\widehat{X}},$$

so $\gamma^{(q)}$ is also causal.

Proof. With probability one, each $\widehat{x}$ in the support of $\widehat{\mathbb{P}}$ corresponds to only one $\widehat{z}$, so that

$$\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}=\widehat{x}_k} = \delta_{\widehat{z}_k}.$$

Now let $\gamma \in \Gamma(\widehat{\mathbb{P}}, \mathbb{P})$. Because

$$\mathbb{E}_{X|\widehat{X}}[\gamma_{\widehat{Z}|(\widehat{X},X)}] = \gamma_{\widehat{Z}|\widehat{X}} = \delta_{\widehat{Z}},$$

the only choice is $\gamma_{\widehat{Z}|(\widehat{X},X)} = \delta_{\widehat{X}}$, for $(\gamma_1)_{X|\widehat{X}}$-a.e. X. Therefore, γ is causal. □

EC.2. Supremum of Convex Functions

In this subsection, we provide several auxiliary results on the properties of the supremum of a family of convex functions. Analysis in this subsection will be used in the proof of Theorem 1.

LEMMA EC.2 **(Dual Objective Function)**. *Recall the dual objective function $h(\lambda)$ defined in* (5). *Let $\mathcal{I} = \{\lambda \in [0,\infty) : h(\lambda) < \infty\}$. Then it has the following properties:*

(I) *There exists $\kappa \geq 0$, such that either $\mathcal{I} = (\kappa, \infty)$ or $\mathcal{I} = [\kappa, \infty)$.*

(II) *h is convex and continuous in $\mathcal{I}$.*

(III) *$h(\lambda) \to \infty$ as $\lambda \to \infty$.*

(IV) *h has a minimizer $\lambda^* \in [\kappa, \infty)$.*

Proof. (I) $h(\lambda) - \lambda\rho^p$ is monotonically decreasing in λ, therefore we can find κ such that h is infinite for smaller λ, and finite for greater λ.

(II) h is a combination of supremums and expectations of convex functions, and therefore h is convex. Since $h < \infty$ in $\mathcal{I}$, h is continuous in $\mathcal{I}$ with only a possible exception at $\kappa \in \mathcal{I}$. Now we verify its lower semi-continuity at κ.

$$\begin{aligned}
\liminf_{\lambda\downarrow\kappa} h(\lambda) &= \kappa\rho^p + \liminf_{\lambda\downarrow\kappa} \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\Psi(f(x),z) - \lambda\|z-\widehat{Z}\|^p\right\} \mid \widehat{X}\right] - \lambda\|x-\widehat{X}\|^p\right\}\right] \\
&\geq \kappa\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\liminf_{\lambda\downarrow\kappa}\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\Psi(f(x),z) - \lambda\|z-\widehat{Z}\|^p\right\} \mid \widehat{X}\right] - \kappa\|x-\widehat{X}\|^p\right\}\right] \\
&\geq \kappa\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\liminf_{\lambda\downarrow\kappa}\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\Psi(f(x),z) - \lambda\|z-\widehat{Z}\|^p\right\} \mid \widehat{X}\right] - \kappa\|x-\widehat{X}\|^p\right\}\right] \\
&\geq \kappa\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\liminf_{\lambda\downarrow\kappa}\sup_{z\in\mathcal{Z}}\left\{\Psi(f(x),z) - \lambda\|z-\widehat{Z}\|^p\right\} \mid \widehat{X}\right] - \kappa\|x-\widehat{X}\|^p\right\}\right] \\
&\geq \kappa\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\liminf_{\lambda\downarrow\kappa}\Psi(f(x),z) - \lambda\|z-\widehat{Z}\|^p\right\} \mid \widehat{X}\right] - \kappa\|x-\widehat{X}\|^p\right\}\right] \\
&= \kappa\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\Psi(f(x),z) - \kappa\|z-\widehat{Z}\|^p\right\} \mid \widehat{X}\right] - \kappa\|x-\widehat{X}\|^p\right\}\right] = h(\kappa).
\end{aligned}$$

We repeatedly used the Fatou's lemma and that minimax is greater than maximin. By convexity, h is continuous in $\mathcal{I}$.

(III) This is simply because we can pick $x = \widehat{X}$, $z = \widehat{Z}$ so

$$\begin{aligned}
h(\lambda) &\geq \lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(f(\widehat{X}),\widehat{Z}) - \lambda\|\widehat{Z}-\widehat{Z}\|^p \mid \widehat{X}\right] - \lambda\|\widehat{X}-\widehat{X}\|^p\right] \\
&= \lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(f(\widehat{X}),\widehat{Z}) \mid \widehat{X}\right]\right] = \lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \to +\infty
\end{aligned}$$

as $\lambda \to +\infty$.

(IV) It follows from (I)-(III). □

LEMMA EC.3 **(Exchange Sup and Derivative for Convex Functions)**. *Let Λ be an index set. Let $\{F_\alpha\}_{\alpha\in\Lambda}$ be a family of real-valued convex functions defined on an interval $\mathcal{I}$. Suppose its sup is pointwise bounded, $\Phi(\lambda) = \sup_{\alpha\in\Lambda} F_\alpha(\lambda) < \infty$. Denote $f_\alpha(\lambda) = F'_\alpha(\lambda)$, and $\phi(\lambda) = \Phi'(\lambda)$. For any function f, we denote f^* [resp. f_*] to be the upper [resp. lower] semicontinuous envelope of f. For every $\varepsilon > 0$, define the ε-argmax set Ω_ε and $\overline{D}, \underline{D}$ by*

$$\Omega_\varepsilon(\lambda) := \{\alpha \in \Lambda : \ F_\alpha(\lambda) \geq \Phi(\lambda) - \varepsilon\},$$
$$\overline{D}_\varepsilon(\lambda) := \sup_{\alpha\in\Omega_\varepsilon(\lambda)} f^*_\alpha(\lambda), \qquad \overline{D}(\lambda) = \lim_{\varepsilon\to 0} \overline{D}_\varepsilon(\lambda),$$
$$\underline{D}_\varepsilon(\lambda) := \inf_{\alpha\in\Omega_\varepsilon(\lambda)} f_{\alpha *}(\lambda), \qquad \underline{D}(\lambda) = \lim_{\varepsilon\to 0} \underline{D}_\varepsilon(\lambda).$$

Then

(I) *For every $\lambda \in \mathcal{I}$, $\underline{D}(\lambda) \leq \overline{D}(\lambda)$.*

(II) *For every $\lambda, \mu \in \mathcal{I}$ with $\lambda < \mu$, $\overline{D}(\lambda) \leq \phi^*(\lambda) \leq \phi_*(\mu) \leq \underline{D}(\mu)$.*

(III) *Fix $\lambda \in \mathcal{I}$, $\delta > 0$, $\epsilon > 0$. If $\lambda_1 \in \mathcal{I}$ such that $\lambda_1 < \lambda$ is sufficiently close to λ, then we can find $\alpha \in \Lambda$ such that*

$$f^*_\alpha(\lambda_1) \leq \phi_*(\lambda) + \delta, \qquad F_\alpha(\lambda_1) \geq \Phi(\lambda) - \epsilon.$$

If $\lambda_2 \in \mathcal{I}$ such that $\lambda_2 > \lambda$ is sufficiently close to λ, we can find $\beta \in \Lambda$ such that

$$f_{\beta *}(\lambda_2) \geq \phi^*(\lambda) - \delta, \qquad F_\beta(\lambda_2) \geq \Phi(\lambda) - \epsilon.$$

Proof. Φ is the sup of a family of convex functions, so Φ is convex. Since Φ and F_α are convex and finite in $\mathcal{I}$, they have locally Lipschitz, monotonously increasing derivatives ϕ and f_α. Monotonicity implies f^*_α and ϕ^* [resp. $f_{\alpha *}$ and ϕ_*] are right [resp. left] continuous, and thus convexity implies for $\lambda < \mu$,

$$f_\alpha{}^*(\lambda) \leq \frac{F_\alpha(\mu) - F_\alpha(\lambda)}{\mu - \lambda} \leq f_{\alpha *}(\mu), \qquad \phi^*(\lambda) \leq \frac{\Phi(\mu) - \Phi(\lambda)}{\mu - \lambda} \leq \phi_*(\mu). \tag{EC.1}$$

(I) ε-argmax set Ω_ε is never empty by definition. Therefore, $\underline{D}_\varepsilon(\lambda) \leq \overline{D}_\varepsilon(\lambda)$ holds for all ε. As $\varepsilon \to 0$, $\Omega_\varepsilon(\lambda)$ shrinks, so $\overline{D}_\varepsilon(\lambda) \downarrow \overline{D}(\lambda)$, $\underline{D}_\varepsilon(\lambda) \uparrow \underline{D}(\lambda)$, we have $\underline{D}(\lambda) \leq \overline{D}(\lambda)$.

(II) Fix any $\varepsilon > 0$, and $\lambda < \mu$. For any $\alpha \in \Omega_\varepsilon(\lambda)$, $\beta \in \Omega_\varepsilon(\mu)$, using (EC.1) we have

$$F_\alpha(\mu) - \varepsilon \le \Phi(\mu) - \varepsilon \le F_\beta(\mu) \le F_\beta(\lambda) + (\mu - \lambda) f_{\beta_*}(\mu) \le \Phi(\lambda) + (\mu - \lambda) f_{\beta_*}(\mu),$$
$$F_\beta(\lambda) - \varepsilon \le \Phi(\lambda) - \varepsilon \le F_\alpha(\lambda) \le F_\alpha(\mu) - (\mu - \lambda) {f_\alpha}^*(\lambda) \le \Phi(\mu) - (\mu - \lambda) {f_\alpha}^*(\lambda).$$

By these two inequalities, we conclude

$$-\varepsilon + (\mu - \lambda) f_\alpha^*(\lambda) \le \Phi(\mu) - \Phi(\lambda) \le \varepsilon + (\mu - \lambda) f_{\beta_*}(\mu),$$
$$\Rightarrow -\frac{\varepsilon}{\mu - \lambda} + f_\alpha^*(\lambda) \le \frac{\Phi(\mu) - \Phi(\lambda)}{\mu - \lambda} \le \frac{\varepsilon}{\mu - \lambda} + f_{\beta_*}(\mu).$$

By taking the sup over $\alpha \in \Omega_\varepsilon(\lambda)$, taking the inf over $\beta \in \Omega_\varepsilon(\mu)$, we have

$$-\frac{\varepsilon}{\mu - \lambda} + \overline{D}_\varepsilon(\lambda) \le \frac{\Phi(\mu) - \Phi(\lambda)}{\mu - \lambda} \le \frac{\varepsilon}{\mu - \lambda} + \underline{D}_\varepsilon(\mu).$$

Let $\varepsilon \to 0$,

$$\overline{D}(\lambda) \le \frac{\Phi(\mu) - \Phi(\lambda)}{\mu - \lambda} \le \underline{D}(\mu). \tag{EC.2}$$

We now combine (EC.1) with (EC.2) to show that $\phi^*(\lambda) \le \underline{D}(\mu)$, $\overline{D}(\lambda) \le \phi_*(\mu)$. To finish the proof of (II), we use the monotonicity $\phi^*(\lambda) \le \phi_*(\mu)$, and

$$\phi^*(\lambda) = \lim_{\mu \downarrow \lambda} \phi(\mu) \ge \lim_{\mu \downarrow \lambda} \phi_*(\mu) \ge \overline{D}(\lambda), \qquad \phi_*(\mu) = \lim_{\lambda \uparrow \mu} \phi(\lambda) \le \lim_{\lambda \uparrow \mu} \phi^*(\lambda) \le \underline{D}(\mu).$$

(III) Since Φ is continuous in the interior of $\mathcal{I}$, we can let λ_1 and λ_2 be close enough to λ such that

$$\Phi(\lambda_1), \Phi(\lambda_2) \ge \Phi(\lambda) - \frac{\epsilon}{2}.$$

Let $\varepsilon < \frac{\epsilon}{2}$ be small enough such that $\overline{D}_\varepsilon(\lambda_1) < \overline{D}(\lambda_1) + \delta$, $\underline{D}_\varepsilon(\lambda_2) > \underline{D}(\lambda_2) - \delta$. Pick any $\alpha \in \Omega_\varepsilon(\lambda_1), \beta \in \Omega_\varepsilon(\lambda_2)$, then

$$f_\alpha^*(\lambda_1) \le \overline{D}_\varepsilon(\lambda_1) < \overline{D}(\lambda_1) + \delta \le \phi_*(\lambda) + \delta,$$
$$f_{\beta_*}(\lambda_2) \ge \underline{D}_\varepsilon(\lambda_2) > \underline{D}(\lambda_2) - \delta \ge \phi^*(\lambda) - \delta.$$

Moreover, by the definition of $\Omega_\varepsilon(\lambda)$,

$$F_\alpha(\lambda_1) \ge \Phi(\lambda_1) - \varepsilon \ge \Phi(\lambda_1) - \frac{\epsilon}{2} \ge \Phi(\lambda) - \epsilon,$$
$$F_\beta(\lambda_2) \ge \Phi(\lambda_2) - \varepsilon \ge \Phi(\lambda_2) - \frac{\epsilon}{2} \ge \Phi(\lambda) - \epsilon. \qquad \square$$

LEMMA EC.4. *With the same notations as the previous lemma, let Λ be an Euclidean space. Suppose for each $\lambda \in \mathrm{Int}(\mathcal{I})$, $F_\alpha(\lambda)$ is upper semicontinuous in α, and $|f_\alpha(\lambda)| \to \infty$ as $|\alpha| \to \infty$. Then*

(I) *$\Omega_0(\lambda)$ is nonempty.*

(II) *There exists $\alpha, \beta \in \Omega_0(\lambda)$, such that*

$$f_{\alpha *}(\lambda) = \phi_*(\lambda), \qquad f^*_\beta(\lambda) = \phi^*(\lambda), \qquad F_\alpha(\lambda) = F_\beta(\lambda) = \Phi(\lambda).$$

(III) *$\overline{D}_0(\lambda) = \overline{D}(\lambda) = \phi^*(\lambda)$, and $\underline{D}_0(\lambda) = \underline{D}(\lambda) = \phi_*(\lambda)$.*

Proof. Let $\lambda_0 \in \mathrm{Int}(\mathcal{I})$. Then we can find $\kappa < \lambda_0 < \mu$ all inside $\mathrm{Int}(\mathcal{I})$. For some small δ, $\kappa' = \kappa - \delta$ and $\mu' = \mu + \delta$ are also inside $\mathrm{Int}(\mathcal{I})$.

(I) By Lemma EC.3 (II), $\phi_*(\lambda) \le \underline{D}(\lambda) \le \overline{D}(\lambda) \le \phi^*(\lambda)$, and since λ is in the interior of $\mathcal{I}$, Φ is locally Lipschitz, $\underline{D}(\lambda), \overline{D}(\lambda)$ are finite. Thus for some small ε, $\underline{D}_\varepsilon(\lambda)$ and $\overline{D}_\varepsilon(\lambda)$ are finite. This implies that Ω_ε is bounded, otherwise $|f_\alpha(\lambda)| \to \infty$ as $\alpha \to \infty$. Because F_α is upper semicontinuous, Ω_ε is also closed, so it is compact, thus

$$\Phi(\lambda) = \sup_{\alpha \in \Lambda} F_\alpha(\lambda) = \sup_{\alpha \in \Omega_\varepsilon(\lambda)} F_\alpha(\lambda)$$

is attainable, i.e.,

$$\Omega_0(\lambda) = \arg\max_{\alpha \in \Lambda} F_\alpha(\lambda)$$

is nonempty.

(II) For every λ, since $\Omega_0(\lambda) \subset \Omega_\varepsilon(\lambda)$ for any ε, we know that $\overline{D}_\varepsilon(\lambda) \ge \overline{D}_0(\lambda)$, $\underline{D}_\varepsilon(\lambda) \le \underline{D}_0(\lambda)$. Let $\varepsilon \to 0$ we have $\overline{D}(\lambda) \ge \overline{D}_0(\lambda)$, $\underline{D}(\lambda) \le \underline{D}_0(\lambda)$. So for every $\alpha \in \Omega_0(\lambda)$,

$$\phi_*(\lambda) \le \underline{D}(\lambda) \le \underline{D}_0(\lambda) \le f_{\alpha *}(\lambda) \le f^*_\alpha(\lambda) \le \overline{D}_0(\lambda) \le \overline{D}(\lambda) \le \phi^*(\lambda). \tag{EC.3}$$

Let $\lambda_n \uparrow \lambda_0$ be an increasing sequence inside $[\kappa, \mu]$. For each λ_n, $\Omega_0(\lambda_n)$ is nonempty, so we can find α_n such that

$$F_{\alpha_n}(\lambda_n) = \Phi(\lambda_n), \qquad \phi_*(\lambda_n) \le f_{\alpha_n *}(\lambda_n) \le f^*_{\alpha_n}(\lambda_n) \le \phi^*(\lambda_n).$$

First, we claim that F_{α_n} are uniformly bounded in $[\kappa, \mu]$. The upper bound $F_{\alpha_n} \le \Phi$ is clear. As for the lower bound, we first use the convexity of Φ, for all $\lambda \in [\kappa, \mu]$,

$$\Phi(\lambda) \ge \Phi(\kappa) + \phi^*(\kappa)(\lambda - \kappa), \qquad \Phi(\lambda) \ge \Phi(\mu) - \phi_*(\mu)(\mu - \lambda).$$

then we use the convexity of F_{α_n}, for $\lambda \in [\lambda_n, \mu]$,

$$\begin{aligned} F_{\alpha_n}(\lambda) &\geq F_{\alpha_n}(\lambda_n) + f^*_{\alpha_n}(\lambda_n)(\lambda - \lambda_n) \\ &\geq \Phi(\lambda_n) + \phi_*(\lambda_n)(\lambda - \lambda_n) \\ &\geq \Phi(\kappa) + \phi^*(\kappa)(\lambda_n - \kappa) + \phi^*(\kappa)(\lambda - \lambda_n) \\ &= \Phi(\kappa) + \phi^*(\kappa)(\lambda - \kappa). \end{aligned}$$

For $\lambda \in [\kappa, \lambda_n]$,

$$\begin{aligned} F_{\alpha_n}(\lambda) &\geq F_{\alpha_n}(\lambda_n) - f_{\alpha_n *}(\lambda_n)(\lambda_n - \lambda) \\ &\geq \Phi(\lambda_n) - \phi^*(\lambda_n)(\lambda_n - \lambda) \\ &\geq \Phi(\mu) - \phi_*(\mu)(\mu - \lambda_n) - \phi_*(\mu)(\lambda_n - \lambda) \\ &= \Phi(\mu) - \phi_*(\mu)(\mu - \lambda). \end{aligned} \tag{EC.4}$$

Therefore, for all $\lambda \in [\kappa, \mu]$,

$$F_{\alpha_n}(\lambda) \geq \min\left\{\Phi(\kappa) + \phi^*(\kappa)(\lambda - \kappa), \Phi(\mu) - \phi_*(\mu)(\mu - \lambda)\right\}.$$

Next, we claim that F_{α_n} are equicontinuous in $[\kappa, \mu]$. Since

$$F_{\alpha_n}(\kappa) \geq \min\left\{\Phi(\kappa), \Phi(\mu) - \phi_*(\mu)(\mu - \kappa)\right\} = \Phi(\mu) - \phi_*(\mu)(\mu - \kappa),$$

by convexity of F_{α_n} we have

$$f_{\alpha_n *}(\kappa) \geq \frac{F_{\alpha_n}(\kappa) - F_{\alpha_n}(\kappa')}{\kappa - \kappa'} \geq \frac{\Phi(\mu) - \phi_*(\mu)(\mu - \kappa) - \Phi(\kappa')}{\delta}.$$

Similarly, we have

$$f^*_{\alpha_n}(\mu) \leq \frac{F_{\alpha_n}(\mu') - F_{\alpha_n}(\mu)}{\mu' - \mu} \leq \frac{\Phi(\mu') - \Phi(\kappa) - \phi^*(\kappa)(\mu - \kappa)}{\delta}.$$

f_{α_n} are increasing between κ and μ, so they are uniformly bounded, thus F_{α_n} are uniformly Lipschitz.

Since f_{α_n} are uniformly bounded, we know that $\{\alpha_n\}_{n\in\mathbb{N}}$ is bounded by the assumption of the lemma. Up to a subsequence, we may assume $\alpha_n \to \alpha$. Since F_{α_n} are uniformly bounded and equicontinuous in $[\kappa, \mu]$, by Arzelà–Ascoli Lemma it admits a subsequence uniformly

converging to some F_∞, and since F_α is upper semicontinuous in α, we know that $F_\alpha \geq \lim_{n\to\infty} F_{\alpha_n} = F_\infty$. Therefore, up to a subsequence,

$$\Phi(\lambda_0) \geq F_\alpha(\lambda_0) \geq F_\infty(\lambda_0) = \lim_{n\to\infty} F_{\alpha_n}(\lambda_n) = \lim_{n\to\infty} \Phi(\lambda_n) = \Phi(\lambda_0).$$

Thus $\alpha \in \Omega_0(\lambda_0)$. Moreover, by taking $n \to \infty$ in (EC.4), for any $\lambda \in [\kappa, \lambda_0)$ we have

$$\begin{aligned}\Phi(\lambda) \geq F_\alpha(\lambda) \geq F_\infty(\lambda) &= \lim_{n\to\infty} F_{\alpha_n}(\lambda_n) - f_{\alpha_n *}(\lambda_n)(\lambda_n - \lambda) \\ &\geq \lim_{n\to\infty} F_{\alpha_n}(\lambda_n) - \phi_*(\lambda_n)(\lambda_n - \lambda) = \Phi(\lambda_0) - \phi_*(\lambda_0)(\lambda_0 - \lambda),\end{aligned}$$

and they all equal at $\lambda = \lambda_0$. So the left derivative at λ_0

$$\phi_*(\lambda_0) \geq f_{\alpha *}(\lambda_0) \geq \phi_*(\lambda_0)$$

are equal. This shows that $f_{\alpha *}(\lambda_0) = \phi_*(\lambda_0)$. The proof for the β part is exactly symmetric to the α, so we omit it here.

(III) This is the consequence of part (II) and (EC.3). □

The proofs in this appendix were inspired by the construction in Gao and Kleywegt (2023). However, because of the two-layer structure of the causal transport distance, we need Lemma EC.3 and EC.4, which cover more general scenarios than the ones considered in Gao and Kleywegt (2023).

EC.3. Proofs for Section 3.1

Proof of Theorem 1. It suffices to prove the direction $v_{\mathsf{P}}^f \geq v_{\mathsf{D}}^f$. We may assume

$$\mathbb{E}_{\widehat{\mathbb{P}}}[\Psi(f(\widehat{X}), \widehat{Z})] < \infty, \tag{EC.5}$$

otherwise $v_{\mathsf{P}}^f = \infty$ because $\widehat{\mathbb{P}}$ is feasible, and the strong duality holds automatically.

For each $x \in \mathcal{X}$, $\widehat{z} \in \mathcal{Z}$ we denote

$$G_{(z)}(\lambda; x, \widehat{z}) := \Psi(f(x), z) - \lambda \|z - \widehat{z}\|^p.$$

It is a linearly decreasing function of λ. Thus, the supremum over z

$$\Upsilon(\lambda; x, \widehat{z}) := \sup_{z\in\mathcal{Z}} \left\{G_{(z)}(\lambda; x, \widehat{z})\right\} \tag{EC.6}$$

is a decreasing convex function of λ. Because the expectation of decreasing convex functions are decreasing and convex, we have for each $\widehat{x} \in \mathcal{X}$,

$$F_{(x)}(\lambda;\widehat{x}) := \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;x,\widehat{Z}) \mid \widehat{X}=\widehat{x}\right] - \lambda\|x-\widehat{x}\|^p$$

is a family of decreasing convex functions of λ. Their supremum

$$\Phi(\lambda;\widehat{x}) := \sup_{x\in\mathcal{X}}\left\{F_{(x)}(\lambda;\widehat{x})\right\} \tag{EC.7}$$

is again convex and decreasing. Finally, the dual objective function

$$h(\lambda) = \lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda;\widehat{X})\right]$$

is also convex. By Lemma EC.2, there exists $\kappa \in [0,\infty]$ such that h is finite in (κ,∞) and infinite in $[0,\kappa)$. Moreover, in the case $\kappa < \infty$, h attains its global minimum at $\lambda^* \geq \kappa$. Therefore, we can separate the following cases.

Case 1: $\kappa = \infty$

This means $h(\lambda) = \infty$ for any $\lambda \geq 0$, therefore $v_{\mathsf{D}}^f = \infty$. Now fix $\lambda > 0$, then $h(\lambda) = 0$ implies

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda;\widehat{X})\right] = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}} F_{(x)}(\lambda;\widehat{X})\right] = \infty.$$

Recall that $\widehat{\mathbb{P}}_{\widehat{X}}$ is a discrete measure. Therefore, for some $\widehat{x}_*$ in the support of $\widehat{\mathbb{P}}_{\widehat{X}}$, $\Phi(\lambda;\widehat{x}_*) = \infty$. For some large number $N > 0$ to be determined, by the definition of $\Phi(\lambda;\widehat{x})$, one can find $x_* \in \mathcal{X}$ such that $F_{(x_*)}(\lambda;\widehat{x}_*) > N$. Define $T_1 : \mathcal{X} \to \mathcal{X}$ by

$$T_1(x) = \begin{cases} x & x \neq \widehat{x}_* \\ x_* & x = \widehat{x}_* \end{cases}.$$

Then

$$\begin{aligned}\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[F_{(T_1(\widehat{X}))}(\lambda;\widehat{X})\right] &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[F_{(\widehat{X})}(\lambda;\widehat{X})\mathbf{1}\{\widehat{X}\neq\widehat{x}_*\}\right] + \widehat{\mathbb{P}}_{\widehat{X}}(\{\widehat{x}_*\})\cdot F_{(x_*)}(\lambda;\widehat{x}_*) \\ &\geq \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[F_{(\widehat{X})}(\lambda;\widehat{X})\mathbf{1}\{\widehat{X}\neq\widehat{x}_*\}\right] + \widehat{\mathbb{P}}_{\widehat{X}}(\{\widehat{x}_*\})\cdot N\end{aligned}$$

can be made arbitrarily large by selecting sufficiently large N. In particular, we can choose N large enough to ensure

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[F_{T_1(\widehat{X})}(\lambda;\widehat{X})\right] \geq \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] + 2\lambda\rho^p.$$

Here we have used the assumption (EC.5). We may abbreviate as $X := T_1(\widehat{X})$, so that

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[F_X(\lambda;\widehat{X})\right] \geq \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] + 2\lambda\rho^p.$$

By the definition of $F_{(x)}$, we have

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;X,\widehat{Z}) \mid \widehat{X}\right] - \lambda\|X-\widehat{X}\|^p\right] \geq \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] + 2\lambda\rho^p.$$

Rearranging the terms, we conclude

$$\begin{aligned} 2\lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\lambda\|X-\widehat{X}\|^p\right] &\leq \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;X,\widehat{Z}) - \Psi(f(\widehat{X}),\widehat{Z}) \mid \widehat{X}\right]\right] \\ &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}} G_{(z)}(\lambda;X,\widehat{Z}) - \Psi(f(\widehat{X}),\widehat{Z}) \mid \widehat{X}\right]\right] \end{aligned} \tag{EC.8}$$

We would like to construct $T_2 : \mathcal{X}\times\mathcal{Z} \to \mathcal{Z}$ in two different scenarios. Fix $\widehat{x}_k$ in the support of $\widehat{\mathbb{P}}_{\widehat{X}}$ and $x_k = T_1(\widehat{x}_k)$. The first scenario would be if $\Upsilon(\lambda;x_k,\widehat{z}_*) = \infty$ for some $\widehat{z}_*$ within the support of $\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}=\widehat{x}_k}$. Similar to the construction of T_1, one can find some one-point transport map $T_2(\widehat{x}_k,\cdot)$ such that

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[G_{(T_2(\widehat{x}_k,\widehat{Z}))}(\lambda;x_k,\widehat{Z}) \mid \widehat{X}=\widehat{x}_k\right] \geq N$$

for any large N to be prescribed. The second scenario would be if $\Upsilon(\lambda;x_k,\widehat{z}_*) < \infty$ for every $\widehat{z}_*$ within the support of $\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}$. For each $(\widehat{x}_k,\widehat{z}_{ki})$ pair, we can find $z_{ki}\in\mathcal{Z}$ such that

$$G_{(z_{ki})}(\lambda;x_k,\widehat{z}_{ki}) \geq \Upsilon(\lambda;x_k,\widehat{z}_{ki}) - \lambda\rho^p.$$

Define $T_2(x_k,\widehat{z}_{ki}) = z_{ki}$, then

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[G_{(T_2(\widehat{x}_k,\widehat{Z}))}(\lambda;x_k,\widehat{Z}) \mid \widehat{X}=\widehat{x}_k\right] \geq \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;x_k,\widehat{Z}) \mid \widehat{X}=\widehat{x}_k\right] - \lambda\rho^p.$$

Combine two scenarios and denote $Z = T_2(\widehat{X},\widehat{Z})$, then we have

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[G_{(Z)}(\lambda;X,\widehat{Z}) \mid \widehat{X}\right]\right] \geq \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;X,\widehat{Z}) \mid \widehat{X}\right]\right] - \lambda\rho^p.$$

Indeed, if the first scenario happens for any $\widehat{x}_k$, then the right-hand side can be made arbitrarily large. Together with (EC.8), we have

$$\begin{aligned} \lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\lambda\|X-\widehat{X}\|^p\right] &\leq \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[G_{(Z)}(\lambda;X,\widehat{Z}) - \Psi(f(\widehat{X}),\widehat{Z}) \mid \widehat{X}\right]\right] \\ &= \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(X),Z) - \Psi(f(\widehat{X}),\widehat{Z}) - \lambda\|Z-\widehat{Z}\|^p\right]. \end{aligned}$$

Assemble T_1 and T_2 yields a causal transport map $T : \mathcal{X} \times \mathcal{Z} \to \mathcal{X} \times \mathcal{Z}$ defined by $T(\widehat{x}, \widehat{z}) = (T_1(\widehat{x}), T_2(\widehat{x}, \widehat{z}))$. It induces a causal transport plan $\gamma_1 = (T \otimes \mathrm{id}_{\mathcal{X}\times\mathcal{Z}})_\sharp \widehat{\mathbb{P}}$, with $\sharp$ denotes push-forward of a measure. Then $((X, Z), (\widehat{X}, \widehat{Z})) \sim \gamma_1$. Denote the transportation cost between $(\widehat{X}, \widehat{Z})$ and (X, Z) via γ_1 by

$$D = \mathbb{E}_{\gamma_1}\left[\|X - \widehat{X}\|^p + \|Z - \widehat{Z}\|^p\right],$$

then

$$\mathbb{E}_{\gamma_1}\left[\Psi(f(X), Z) - \Psi(f(\widehat{X}), \widehat{Z})\right] \geq \lambda\rho^p + \lambda D.$$

Let $\gamma_0 = (\mathrm{id}_{\mathcal{X}\times\mathcal{Z}} \otimes \mathrm{id}_{\mathcal{X}\times\mathcal{Z}})_\sharp \widehat{\mathbb{P}}$ denote the joint distribution induced by identity transport map. Let $\gamma_\theta = \theta\gamma_1 + (1-\theta)\gamma_0$ be the transport plan which perturbs γ_0 by moving $\theta := \min\{1, \frac{\rho^p}{D}\}$ portion of mass from $(\widehat{X}, \widehat{Z})$ to (X, Z). By the convexity lemma 1, this transport plan is causal. Denote $\mathbb{P}_\theta = (\gamma_\theta)_{(X,Z)}$ to be the marginal of γ_θ. Then

$$\mathsf{C}_p(\widehat{\mathbb{P}}, \mathbb{P})^p \leq \mathbb{E}_{\gamma_\theta}\left[\|X - \widehat{X}\|^p + \|Z - \widehat{Z}\|^p\right] = \theta D \leq \rho^p,$$

So $\mathbb{P}_\theta$ is primal feasible, and

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}_\theta}[\Psi(f(X), Z)] - \mathbb{E}_{\widehat{\mathbb{P}}}[\Psi(f(\widehat{X}), \widehat{Z})] &= \mathbb{E}_{\gamma_\theta}\left[\Psi(f(X), Z) - \Psi(f(\widehat{X}), \widehat{Z})\right] \\
&= \theta\mathbb{E}_{\gamma_1}\left[\Psi(f(x), Z) - \Psi(f(\widehat{x}), \widehat{Z})\right] \\
&\geq \theta\left(\lambda\rho^p + \lambda D\right) \\
&\geq \lambda\rho^p.
\end{aligned}$$

Therefore

$$v_{\mathsf{P}}^f \geq \mathbb{E}_{\mathbb{P}_\theta}[\Psi(f(X), Z)] \geq \mathbb{E}_{\widehat{\mathbb{P}}}[\Psi(f(\widehat{X}), \widehat{Z})] + \lambda\rho^p,$$

and since λ can be arbitrarily large, we have

$$v_{\mathsf{P}}^f = \infty = v_{\mathsf{D}}^f.$$

Case 2: $\kappa < \infty, \lambda^* > \kappa$

Fix some small $\delta > 0$, $\varepsilon > 0$. Applying Lemma EC.3 on (EC.7), for $\widehat{x} \in \mathcal{X}$ we can find $\overline{x}, \underline{x} \in \mathcal{X}$ such that

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^+} F_{(\underline{x})}(\lambda_1; \widehat{x}) \le \frac{\mathrm{d}}{\mathrm{d}\lambda^-} \Phi(\lambda^*; \widehat{x}) + \delta, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda^-} F_{(\overline{x})}(\lambda_2; \widehat{x}) \ge \frac{\mathrm{d}}{\mathrm{d}\lambda^+} \Phi(\lambda^*; \widehat{x}) - \delta,$$
$$F_{(\underline{x})}(\lambda_1, \widehat{x}) \ge \Phi(\lambda^*, \widehat{x}) - \varepsilon, \qquad F_{(\overline{x})}(\lambda_2, \widehat{x}) \ge \Phi(\lambda^*, \widehat{x}) - \varepsilon$$

for $\kappa < \lambda_1 < \lambda^* < \lambda_2$ and λ_1, λ_2 sufficiently close to λ^*. Fix $x \in \mathcal{X}$. Apply Lemma EC.3 on (EC.6), for $\widehat{z} \in \mathcal{Z}$ we can find $\overline{z}, \underline{z} \in \mathcal{Z}$ such that

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^+} G_{(\underline{z})}(\lambda_3; x, \widehat{z}) \le \frac{\mathrm{d}}{\mathrm{d}\lambda^-} \Upsilon(\lambda_1; x, \widehat{z}) + \delta, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda^-} G_{(\overline{z})}(\lambda_4; x, \widehat{z}) \ge \frac{\mathrm{d}}{\mathrm{d}\lambda^+} \Upsilon(\lambda_2; x, \widehat{z}) - \delta,$$
$$G_{(\underline{z})}(\lambda_3; x, \widehat{z}) \ge \Upsilon(\lambda_1, x, \widehat{z}) - \varepsilon, \qquad G_{(\overline{z})}(\lambda_4; x, \widehat{z}) \ge \Upsilon(\lambda_2, x, \widehat{z}) - \varepsilon$$

for $\kappa < \lambda_3 < \lambda_1 < \lambda^* < \lambda_2 < \lambda_4$ and λ_3, λ_4 sufficiently close to λ_1, λ_2. Now suppose $\widehat{\mathbb{P}}$ is supported over a finite set of $\{(\widehat{x}_k, \widehat{z}_{ki})\}_{ki}$, we know that for $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ sufficiently close to λ^* we can find $\overline{x}_k, \underline{x}_k, \overline{z}_{ki}, \underline{z}_{ki}$ such that the above are satisfied simultaneously. We denote the transport map by $\overline{x}_k = \overline{T}_1(\widehat{x}_i)$, $\overline{z}_{ki} = \overline{T}_2(\widehat{x}_k, \widehat{z}_{ki})$, and $\overline{T}(\widehat{x}_k, \widehat{z}_{ki}) = (\overline{x}_k, \overline{z}_{ki})$. We define $\underline{T}$ similarly, so we can construct $(\overline{X}, \overline{Z}) = \overline{T}(\widehat{X}, \widehat{Z})$, $(\underline{X}, \underline{Z}) = \underline{T}(\widehat{X}, \widehat{Z})$. We denote the law of $((\overline{X}, \overline{Z}), (\underline{X}, \underline{Z}))$ by $\overline{\gamma} = (\overline{T} \otimes \mathrm{id}_{\mathcal{X}\times\mathcal{Z}})_\sharp \widehat{\mathbb{P}}$, and the law of $(\overline{X}, \overline{Z})$ is $\overline{\mathbb{P}} = \overline{\gamma}_{(X,Z)}$ the marginal. Similarly we define $\underline{\gamma}$ and $\underline{\mathbb{P}}$. We also define $\widehat{\gamma} = (\mathrm{id}_{\mathcal{X}\times\mathcal{Z}} \otimes \mathrm{id}_{\mathcal{X}\times\mathcal{Z}})_\sharp \widehat{\mathbb{P}}$ to be the identity transport plan. For convenience, denote the law of $(\overline{X}, \widehat{X})$ to be $\overline{\gamma}_1 = \overline{\gamma}_{(X,\widehat{X})}$, and the law of $(\underline{X}, \widehat{X})$ to be $\underline{\gamma}_1 = \underline{\gamma}_{(X,\widehat{X})}$. Similarly define $\overline{\gamma}_2 = \overline{\gamma}_{(Z,\widehat{Z})|(X,\widehat{X})}$ and $\underline{\gamma}_2 = \underline{\gamma}_{(Z,\widehat{Z})|(X,\widehat{X})}$ to be the conditional law of $(\overline{Z}, \widehat{Z})$ and $(\underline{Z}, \widehat{Z})$ given $(\overline{X}, \widehat{X})$ and $(\underline{X}, \widehat{X})$, respectively.

We know that $h(\lambda)$ attains its minimum v_{D}^f at some $\lambda^* \in \mathcal{I}$, so $h'(\lambda^*+) \ge 0$ and $h'(\lambda^*-) \le 0$ (if $\lambda^* > \kappa$), so

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda^*} \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda, \widehat{X})\right] \le -\rho^p \le \frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda^*} \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda, \widehat{X})\right]$$

where

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda^*} \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda, \widehat{X})\right] &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda^*} \Phi(\lambda, \widehat{X})\right] \\
&\ge \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda_1} F_{(\underline{X})}(\lambda; \widehat{X})\right] - \delta \\
&= \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda_1} \left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda; \underline{X}, \widehat{Z}) \mid (\underline{X}, \widehat{X})\right] - \lambda\|\underline{X} - \widehat{X}\|^p\right\}\right] - \delta
\end{aligned}$$

$$
\begin{aligned}
&= \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda_1}\Upsilon(\lambda;\underline{X},\widehat{Z})\mid(\underline{X},\widehat{X})\right]-\|\underline{X}-\widehat{X}\|^p\right]-\delta\\
&\geq \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda_3}G_{(\underline{Z})}(\lambda;\underline{X},\widehat{Z})\mid(\underline{X},\widehat{X})\right]-\|\underline{X}-\widehat{X}\|^p\right]-2\delta\\
&\geq \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[-\|\underline{Z}-\widehat{Z}\|^p\mid(\underline{X},\widehat{X})\right]-\|\underline{X}-\widehat{X}\|^p\right]-2\delta\\
&= -\mathbb{E}_{((\underline{X},\underline{Z}),(\widehat{X},\widehat{Z}))\sim\underline{\gamma}}\left[\|\underline{X}-\widehat{X}\|^p+\|\underline{Z}-\widehat{Z}\|^p\right]-2\delta,
\end{aligned}
$$

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda^*}\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda,\widehat{X})\right] &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\bigg|_{\lambda=\lambda^*}\Phi(\lambda,\widehat{X})\right]\\
&\leq \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda_2}F_{(\overline{X})}(\lambda;\widehat{X})\right]+\delta\\
&= \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda_2}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;\overline{X},\widehat{Z})\mid(\overline{X},\widehat{X})\right]-\lambda\|\overline{X}-\widehat{X}\|^p\right\}\right]+\delta\\
&= \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda_2}\Upsilon(\lambda;\overline{X},\widehat{Z})\mid(\overline{X},\widehat{X})\right]-\|\overline{X}-\widehat{X}\|^p\right]+\delta\\
&\leq \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\mathbb{E}_{(\overline{Z},\widehat{Z})\sim\overline{\gamma}_2}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\bigg|_{\lambda=\lambda_4}G_{(\overline{Z})}(\lambda;\overline{X},\widehat{Z})\mid(\overline{X},\widehat{X})\right]-\|\overline{X}-\widehat{X}\|^p\right]+2\delta\\
&\leq \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\mathbb{E}_{(\overline{Z},\widehat{Z})\sim\overline{\gamma}_2}\left[-\|\overline{Z}-\widehat{Z}\|^p\mid(\overline{X},\widehat{X})\right]-\|\overline{X}-\widehat{X}\|^p\right]+2\delta\\
&= -\mathbb{E}_{((\overline{X},\overline{Z}),(\widehat{X},\widehat{Z}))\sim\overline{\gamma}}\left[\|\underline{X}-\widehat{X}\|^p+\|\underline{Z}-\widehat{Z}\|^p\right]+2\delta,
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
\overline{d} &:= \mathbb{E}_{((\overline{X},\overline{Z}),(\widehat{X},\widehat{Z}))\sim\overline{\gamma}}\left[\|\underline{X}-\widehat{X}\|^p+\|\underline{Z}-\widehat{Z}\|^p\right]\leq\rho^p+2\delta,\\
\underline{d} &:= \mathbb{E}_{((\underline{X},\underline{Z}),(\widehat{X},\widehat{Z}))\sim\underline{\gamma}}\left[\|\underline{X}-\widehat{X}\|^p+\|\underline{Z}-\widehat{Z}\|^p\right]\geq\rho^p-2\delta.
\end{aligned}
$$

Based on these, we construct a feasible primal solution. There exists $q_\delta^\varepsilon\in[0,1]$ depending on $\lambda_1,\lambda_2,\lambda_3,\lambda_4$, such that

$$
\begin{aligned}
\rho^p &= (1-q_\delta^\varepsilon)\left(\overline{d}-2\delta\right)+q_\delta^\varepsilon\left(\underline{d}+2\delta\right),\\
\rho^p+2(1-2q_\delta^\varepsilon)\delta &= (1-q_\delta^\varepsilon)\overline{d}+q_\delta^\varepsilon\underline{d}.
\end{aligned}
$$

Let $q^\delta := \frac{\rho^p}{\rho^p+2(1-2q_\delta^\varepsilon)_+\delta}\leq 1$. Define a transport plan $\gamma_\delta^\varepsilon$ by

$$
\gamma_\delta^\varepsilon := q^\delta\left[(1-q_\delta^\varepsilon)\overline{\gamma}+q_\delta^\varepsilon\underline{\gamma}\right]+(1-q^\delta)\widehat{\gamma}.
$$

Its marginal distribution $\mathbb{P}_\delta^\varepsilon = (\gamma_\delta^\varepsilon)_{(X,Z)}$ is given by

$$\mathbb{P}_\delta^\varepsilon = q^\delta \left[(1-q_\delta^\varepsilon)\overline{\mathbb{P}} + q_\delta^\varepsilon \underline{\mathbb{P}}\right] + (1-q^\delta)\widehat{\mathbb{P}}.$$

Then $\mathbb{P}_\delta^\varepsilon$ is primal feasible because

$$\begin{aligned}\mathsf{C}_p(\mathbb{P}_\delta^\varepsilon, \widehat{\mathbb{P}})^p &\le \mathbb{E}_{((X,Z),(\widehat{X},\widehat{Z}))\sim\gamma_\delta^\varepsilon}\left[\|X-\widehat{X}\|^p + \|Z-\widehat{Z}\|^p\right] \\ &\le q^\delta\left[(1-q_\delta^\varepsilon)\overline{d} + q_\delta^\varepsilon \underline{d}\right] \le \rho^p.\end{aligned}$$

In the meantime,

$$\begin{aligned}v_{\mathsf{D}}^f - \lambda^*\rho^p &= h(\lambda^*) - \lambda^*\rho^p \\ &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda^*, \widehat{X})\right] \\ &\le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[F_{(\underline{X})}(\lambda_1;\widehat{X})\right] + \varepsilon \\ &= \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda_1;\underline{X},\widehat{Z}) \mid \widehat{X}\right] - \lambda_1\|\underline{X}-\widehat{X}\|^p\right] + \varepsilon \\ &\le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[G_{(\underline{Z})}(\lambda_3;\underline{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right] - \lambda_1\|\underline{X}-\widehat{X}\|^p\right] + 2\varepsilon \\ &\le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[\Psi(f(\underline{X}),\underline{Z}) - \lambda_3\|\underline{Z}-\widehat{Z}\|^p \mid (\underline{X},\widehat{X})\right] - \lambda_1\|\underline{X}-\widehat{X}\|^p\right] + 2\varepsilon \\ &\le \mathbb{E}_{((\underline{X},\underline{Z}),(\widehat{X},\widehat{Z}))\sim\underline{\gamma}}\left[\Psi(f(\underline{X}),\underline{Z}) - \lambda_3\|\underline{Z}-\widehat{Z}\|^p - \lambda_1\|\underline{X}-\widehat{X}\|^p\right] + 2\varepsilon \\ &\le \mathbb{E}_{\underline{\mathbb{P}}}\left[\Psi(f(\underline{X}),\underline{Z})\right] - \lambda_3\underline{d} + 2\varepsilon.\end{aligned}$$

Similarly

$$\begin{aligned}v_{\mathsf{D}}^f - \lambda^*\rho^p &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda^*, \widehat{X})\right] \\ &\le \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[F_{(\overline{X})}(\lambda_2;\widehat{X})\right] + \varepsilon \\ &= \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda_2;\overline{X},\widehat{Z}) \mid (\overline{X},\widehat{X})\right] - \lambda_2\|\overline{X}-\widehat{X}\|^p\right] + \varepsilon \\ &\le \mathbb{E}_{(\overline{X},\widehat{X})\sim\overline{\gamma}_1}\left[\mathbb{E}_{(\overline{Z},\widehat{Z})\sim\overline{\gamma}_2}\left[G_{(\overline{Z})}(\lambda_4;\overline{X},\widehat{Z}) \mid (\overline{X},\widehat{X})\right] - \lambda_2\|\overline{X}-\widehat{X}\|^p\right] + 2\varepsilon \\ &\le \mathbb{E}_{((\overline{X},\overline{Z}),(\widehat{X},\widehat{Z}))\sim\overline{\gamma}}\left[\Psi(f(\overline{X}),\overline{Z}) - \lambda_4\|\overline{Z}-\widehat{Z}\|^p - \lambda_2\|\overline{X}-\widehat{X}\|^p\right] + 2\varepsilon \\ &\le \mathbb{E}_{\overline{\mathbb{P}}}\left[\Psi(f(\overline{X}),\overline{Z})\right] - \lambda_2\overline{d} + 2\varepsilon.\end{aligned}$$

Therefore,

$$v_{\mathsf{P}}^f \ge \mathbb{E}_{(X,Z)\sim\mathbb{P}_\delta^\varepsilon}\left[\Psi(f(X),Z)\right]$$

$$
\begin{aligned}
&= q^\delta \left((1-q_\delta^\varepsilon)\mathbb{E}_{\overline{\mathbb{P}}}\left[\Psi(f(\overline{X}),\overline{Z})\right] + q_\delta^\varepsilon \mathbb{E}_{\underline{\mathbb{P}}}\left[\Psi(f(\underline{X}),\underline{Z})\right]\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\geq q^\delta \left((1-q_\delta^\varepsilon)\left(v_{\mathsf{D}}^f - \lambda^*\rho^p + \lambda_2\overline{d} - 2\varepsilon\right) + q_\delta^\varepsilon\left(v_{\mathsf{D}}^f - \lambda^*\rho^p + \lambda_3\underline{d} - 2\varepsilon\right)\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\geq q^\delta \left(v_{\mathsf{D}}^f - \lambda^*\rho^p + \lambda_3((1-q_\delta^\varepsilon)\overline{d} + q_\delta^\varepsilon\underline{d}) - 2\varepsilon\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\geq q^\delta \left(v_{\mathsf{D}}^f - \lambda^*\rho^p + \lambda_3(\rho^p + 2(1-2q_\delta^\varepsilon)\delta) - 2\varepsilon\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&= q^\delta \left(v_{\mathsf{D}}^f - (\lambda^*-\lambda_3)\rho^p + 2\lambda_3(1-2q_\delta^\varepsilon)\delta - 2\varepsilon\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right].
\end{aligned}
$$

As $\delta \to 0$, $q^\delta \to 1$. Thus take the limit as $\lambda_3 \to \lambda^*$ and $\delta \to 0$, it follows that

$$v_{\mathsf{P}}^f \geq v_{\mathsf{D}}^f - 2\varepsilon.$$

Since ε can be taken arbitrarily small, $v_{\mathsf{P}}^f \geq v_{\mathsf{D}}^f$.

Case 3: $\lambda^* = \kappa < \infty$

In this case, we can still choose $\overline{x}, \overline{z}$, and we still have

$$F_{(\overline{x})}(\lambda_2, \widehat{x}) > \Phi(\lambda^*, \widehat{x}) - \varepsilon, \qquad G_{(\overline{z})}(\lambda_4; x, \widehat{z}) > \Upsilon(\lambda_2, x, \widehat{z}) - \varepsilon.$$

and

$$\overline{d} = \mathbb{E}_{\overline{\gamma}}\left[\|\overline{X} - \widehat{X}\|^p + \|\overline{Z} - \widehat{Z}\|^p\right] \leq \rho^p + 2\delta.$$

We separate the cases $\kappa = 0$ and $\kappa > 0$.

Case 3.1: $\lambda^* = \kappa = 0$

Let $q^\delta := \frac{\rho^p}{\rho^p + 2\delta} \leq 1$. Define $\gamma_\delta^\varepsilon := q^\delta\overline{\gamma} + (1-q^\delta)\widehat{\gamma}$, then its marginal is a distribution $\mathbb{P}_\delta^\varepsilon$ given by

$$\mathbb{P}_\delta^\varepsilon := q^\delta\overline{\mathbb{P}} + (1-q^\delta)\widehat{\mathbb{P}}.$$

Then it is primal feasible because

$$\mathsf{C}_p(\mathbb{P}_\delta^\varepsilon, \widehat{\mathbb{P}})^p \leq \mathbb{E}_{\gamma_\delta^\varepsilon}\left[\|\overline{X} - \widehat{X}\|^p + \|\overline{Z} - \widehat{Z}\|^p\right] \leq q^\delta\overline{d} \leq \rho^p,$$

thus

$$
\begin{aligned}
v_{\mathsf{P}}^f &\geq \mathbb{E}_{(X,Z)\sim\mathbb{P}_\delta^\varepsilon}\left[\Psi(f(X),Z)\right] \\
&= q^\delta \mathbb{E}_{(\overline{X},\overline{Z})\sim\overline{\mathbb{P}}}\left[\Psi(f(\overline{X}),\overline{Z})\right] + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}[\Psi(f(\widehat{X}),\widehat{Z})] \\
&\geq q^\delta\left(v_{\mathsf{D}}^f - \lambda^*\rho^p + \lambda_2\overline{d} - 2\varepsilon\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}[\Psi(f(\widehat{X}),\widehat{Z})] \\
&\geq q^\delta\left(v_{\mathsf{D}}^f - 2\varepsilon\right) + (1-q^\delta)\mathbb{E}_{\widehat{\mathbb{P}}}[\Psi(f(\widehat{X}),\widehat{Z})]
\end{aligned}
$$

using $\lambda^* = 0$. Let $\delta \to 0$, $q^\delta \to 1$, we have $v_{\mathsf{P}}^f \geq v_{\mathsf{D}}^f - 2\varepsilon$, and by taking $\varepsilon \to 0$ we have $v_{\mathsf{P}}^f \geq v_{\mathsf{D}}^f$.

Case 3.2: $\lambda^* = \kappa > 0$

Fix any $0 < \kappa' < \kappa$. We have

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\kappa';\widehat{X}) - \Phi(\kappa;\widehat{X})\right] = h(\kappa') - h(\kappa) = \infty. \tag{EC.9}$$

We denote

$$\mathcal{X}^*(\lambda;\widehat{x}) := \left\{x \in \mathcal{X} : \ F_{(x)}(\lambda;\widehat{x}) \geq F_{(\widehat{x})}(\lambda;\widehat{x})\right\}.$$

Then $\mathcal{X}^*(\lambda;\widehat{x})$ is nonempty because $\widehat{x} \in \mathcal{X}^*(\lambda;\widehat{x})$. Since

$$\Phi(\kappa';\widehat{x}) = \sup_{x\in\mathcal{X}} F_{(x)}(\kappa';\widehat{x}) = \sup_{x\in\mathcal{X}^*(\kappa';\widehat{x})} F_{(x)}(\kappa';\widehat{x}),$$

we can rewrite (EC.9) as

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}^*(\kappa';\widehat{X})} F_{(x)}(\kappa';\widehat{X}) - \Phi(\kappa;\widehat{X})\right] = \infty.$$

Thus for any fixed $R > 0$, we can pick $\underline{X} = \underline{T}_1(\widehat{X}) \in \mathcal{X}^*(\kappa';\widehat{X})$, which induces $\underline{\gamma}_1$, such that

$$\begin{aligned}
R &< \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[F_{(\underline{X})}(\kappa';\widehat{X}) - \Phi(\kappa;\widehat{X})\right] \\
&\leq \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[F_{(\underline{X})}(\kappa';\widehat{X}) - F_{(\underline{X})}(\kappa;\widehat{X})\right] \\
&= \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa;\underline{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right] + (\kappa-\kappa')\|\underline{X}-\widehat{X}\|^p\right].
\end{aligned} \tag{EC.10}$$

Moreover, because $\underline{X} \in \mathcal{X}^*(\kappa';\widehat{X})$, we have

$$\begin{aligned}
F_{(\widehat{X})}(\kappa';\widehat{X}) &\leq F_{(\underline{X})}(\kappa';\widehat{X}), \\
\kappa'\|\underline{X}-\widehat{X}\|^p &\leq \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa';\widehat{X},\widehat{Z}) \mid \widehat{X}\right], \\
\mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\kappa'\|\underline{X}-\widehat{X}\|^p\right] &\leq \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa';\widehat{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right]\right].
\end{aligned} \tag{EC.11}$$

We denote

$$\mathcal{Z}^*(\lambda;x,\widehat{z}) := \left\{z \in \mathcal{Z} : \ G_{(z)}(\lambda;x,\widehat{z}) \geq G_{(\widehat{z})}(\lambda;x,\widehat{z})\right\}.$$

Then $\mathcal{Z}^*(\lambda;x,\widehat{z})$ is nonempty because $\widehat{z} \in \mathcal{Z}^*(\lambda;x,\widehat{z})$. Since

$$\Upsilon(\kappa';x,\widehat{z}) = \sup_{z\in\mathcal{Z}} G_{(z)}(\kappa';x,\widehat{z}) = \sup_{z\in\mathcal{Z}^*(\kappa';x,\widehat{z})} G_{(z)}(\kappa';x,\widehat{z}),$$

we can rewrite (EC.10) and (EC.11) as

$$R < \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}^*(\kappa';\underline{X},\widehat{Z})} G_{(z)}(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa;\underline{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right] + (\kappa-\kappa')\|\underline{X}-\widehat{X}\|^p\right],$$

$$\mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\kappa'\|\underline{X}-\widehat{X}\|^p\right] \le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}^*(\kappa';\underline{X},\widehat{Z})} G_{(z)}(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa';\widehat{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right]\right].$$

Thus we can pick $\underline{Z} = \underline{T}_2(\widehat{X},\widehat{Z}) \in \mathcal{Z}^*(\kappa';\underline{X},\widehat{Z})$, which induces γ_2, such that

$$\begin{aligned}
R-\varepsilon &< \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[G_{(\underline{Z})}(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa;\underline{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right] + (\kappa-\kappa')\|\underline{X}-\widehat{X}\|^p\right] \\
&\le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[G_{(\underline{Z})}(\kappa';\underline{X},\widehat{Z}) - G_{(\underline{Z})}(\kappa;\underline{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right] + (\kappa-\kappa')\|\underline{X}-\widehat{X}\|^p\right] \\
&= \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[(\kappa-\kappa')\|\underline{Z}-\widehat{Z}\|^p \mid (\underline{X},\widehat{X})\right] + (\kappa-\kappa')\|\underline{X}-\widehat{X}\|^p\right] \\
&= (\kappa-\kappa')\underline{d},
\end{aligned}$$

and simultaneously ensure

$$\begin{aligned}
\mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\kappa'\|\underline{X}-\widehat{X}\|^p\right] - \delta &\le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[G_{(\underline{Z})}(\kappa';\underline{X},\widehat{Z}) - \Upsilon(\kappa';\widehat{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right]\right] \\
&\le \mathbb{E}_{(\underline{X},\widehat{X})\sim\underline{\gamma}_1}\left[\mathbb{E}_{(\underline{Z},\widehat{Z})\sim\underline{\gamma}_2}\left[G_{(\underline{Z})}(\kappa';\underline{X},\widehat{Z}) - G_{(\widehat{Z})}(\kappa';\widehat{X},\widehat{Z}) \mid (\underline{X},\widehat{X})\right]\right] \\
&= \mathbb{E}_{((\underline{X},\underline{Z}),(\widehat{X},\widehat{Z}))\sim\underline{\gamma}}\left[\Psi(f(\underline{X}),\underline{Z}) - \kappa'\|\underline{Z}-\widehat{Z}\|^p - \Psi(f(\widehat{X}),\widehat{Z})\right], \\
\kappa'\underline{d} &\le \mathbb{E}_{((\underline{X},\underline{Z}),(\widehat{X},\widehat{Z}))\sim\underline{\gamma}}\left[\Psi(f(\underline{X}),\underline{Z}) - \Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\le \mathbb{E}_{\underline{\mathbb{P}}}\left[\Psi(f(\underline{X}),\underline{Z})\right] - \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right].
\end{aligned}$$

In conclusion, we have

$$\frac{R-\varepsilon}{\kappa-\kappa'} < \underline{d} \le \frac{\mathbb{E}_{\underline{\mathbb{P}}}\left[\Psi(f(\underline{X}),\underline{Z})\right] - \mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right]}{\kappa'}.$$

We can choose $R = \varepsilon + (\kappa-\kappa')N\rho^p$ for some $N >> 1$ to be specified later. Because

$$\overline{d} - 2\delta \le \rho^p \le \frac{d}{N} \le \underline{d} + 2\delta,$$

there exists $q_\delta^\varepsilon \in [0,1]$ depending on $\lambda_2, \lambda_4, \kappa'$, such that

$$\begin{aligned}
\rho^p &= (1-q_\delta^\varepsilon)\left[\overline{d} - 2\delta\right] + q_\delta^\varepsilon\left[\underline{d} + 2\delta\right], \\
&= (1-q_\delta^\varepsilon)\overline{d} + q_\delta^\varepsilon\underline{d} - 2(1-2q_\delta^\varepsilon)\delta, \\
\rho^p + 2(1-2q_\delta^\varepsilon)\delta &= (1-q_\delta^\varepsilon)\overline{d} + q_\delta^\varepsilon\underline{d}.
\end{aligned}$$

Let $q^\delta := \frac{\rho^p}{\rho^p + 2(1-2q^\varepsilon_\delta)_+\delta} \le 1$. Define a distribution $\mathbb{P}^\varepsilon_\delta$ by

$$\mathbb{P}^\varepsilon_\delta := q^\delta \left[(1-q^\varepsilon_\delta)\overline{\mathbb{P}} + q^\varepsilon_\delta \underline{\mathbb{P}}\right] + \left(1-q^\delta\right)\widehat{\mathbb{P}}.$$

Then $\mathbb{P}^\varepsilon_\delta$ is primal feasible, because

$$\begin{aligned}
\mathsf{C}_p(\mathbb{P}^\varepsilon_\delta, \widehat{\mathbb{P}})^p &\le q^\delta(1-q^\varepsilon_\delta)\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\|\overline{Z}-\widehat{Z}\|^p \mid \widehat{X}\right] + \|\overline{X}-\widehat{X}\|^p\right] \\
&\quad + q^\delta q^\varepsilon_\delta \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\|\underline{Z}-\widehat{Z}\|^p \mid \widehat{X}\right] + \|\underline{X}-\widehat{X}\|^p\right] \\
&\le q^\delta\left[(1-q^\varepsilon_\delta)\overline{d} + q^\varepsilon_\delta \underline{d}\right] \le \rho^p.
\end{aligned}$$

Therefore

$$\begin{aligned}
v^f_{\mathsf{P}} &\ge \mathbb{E}_{(X,Z)\sim\mathbb{P}^\varepsilon_\delta}\left[\Psi(f(X),Z)\right] \\
&= \mathbb{E}_{\overline{\mathbb{P}}}\left[q^\delta(1-q^\varepsilon_\delta)\Psi(f(\overline{X}),\overline{Z})\right] + \mathbb{E}_{\underline{\mathbb{P}}}\left[q^\delta q^\varepsilon_\delta \Psi(f(\underline{X}),\underline{Z})\right] + \mathbb{E}_{\widehat{\mathbb{P}}}\left[(1-q^\delta)\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\ge q^\delta(1-q^\varepsilon_\delta)\left(v^f_{\mathsf{D}} - \kappa\rho^p + \lambda_2\overline{d} - 2\varepsilon\right) + q^\delta q^\varepsilon_\delta \kappa' \underline{d} \\
&\qquad + \left(1-q^\delta + q^\delta q^\varepsilon_\delta\right)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\ge q^\delta\kappa'\left((1-q^\varepsilon_\delta)\overline{d} + q^\varepsilon_\delta\underline{d}\right) + q^\delta(1-q^\varepsilon_\delta)(v^f_{\mathsf{D}} - \kappa\rho^p - 2\varepsilon) \\
&\qquad + \left(1-q^\delta + q^\delta q^\varepsilon_\delta\right)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right] \\
&\ge q^\delta\kappa'(\rho^p + 2(1-2q^\varepsilon_\delta)\delta) + q^\delta(1-q^\varepsilon_\delta)(v^f_{\mathsf{D}} - \kappa\rho^p - 2\varepsilon) + \left(1-q^\delta + q^\delta q^\varepsilon_\delta\right)\mathbb{E}_{\widehat{\mathbb{P}}}\left[\Psi(f(\widehat{X}),\widehat{Z})\right].
\end{aligned}$$

As $\delta \to 0$, we have $q^\delta \to 1$. Moreover, because

$$\rho^p + 2\delta \ge (1-q^\varepsilon_\delta)\overline{d} + q^\varepsilon_\delta\underline{d} \ge q^\varepsilon_\delta\underline{d} \ge q^\varepsilon_\delta N\rho^p,$$

we know that $q^\varepsilon_\delta \le \frac{1+2\delta\rho^{-p}}{N} \to 0$ as $N \to \infty$ and $\delta \to 0$. Therefore, by taking these limits, we have

$$v^f_{\mathsf{P}} \ge \kappa'\rho^p + v^f_{\mathsf{D}} - \kappa\rho^p - 2\varepsilon = v^f_{\mathsf{D}} - 2\varepsilon - (\kappa-\kappa')\rho^p.$$

Since this is true for any $\kappa' < \kappa$ and $\varepsilon > 0$, we may take $\kappa' \to \kappa$ and $\varepsilon \to 0$ so $v^f_{\mathsf{P}} \ge v^f_{\mathsf{D}}$. □

Proof of Theorem 2. Since $\Psi(f(\cdot),\cdot)$ is upper semicontinuous, we know that for each fixed $x \in \mathcal{X}$, $\widehat{z} \in \mathcal{Z}$, $\lambda > \kappa$, $G_{(z)}(\lambda; x, \widehat{z}) = \Psi(f(x), z) - \lambda\|z - \widehat{z}\|^p$ is upper semicontinuous in z. Moreover,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} G_{(z)}(\lambda; x, \widehat{z}) = -\|z-\widehat{z}\|^p \to -\infty \qquad \text{as } |z| \to \infty,$$

By Lemma EC.4 (II), we can find $\overline{z}$, $\underline{z}$ such that

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\Upsilon(\lambda;x,\widehat{z}) = -\|\overline{z}-\widehat{z}\|^p, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda^-}\Upsilon(\lambda;x,\widehat{z}) = -\|\underline{z}-\widehat{z}\|^p, \qquad \Upsilon(\lambda;x,\widehat{z}) = G_{(\overline{z})}(\lambda;x,\widehat{z}) = G_{(\underline{z})}(\lambda;x,\widehat{z}).$$

Now we claim that for each fixed $\widehat{z}\in\mathcal{Z}$, $\lambda>\kappa$, $\Upsilon(\lambda;x,\widehat{z})$ is upper semicontinuous in x. We prove it by contradiction. Assume otherwise, then we can find $x_k\to x$, such that

$$\Upsilon(\lambda;x_k,\widehat{z}) > \Upsilon(\lambda;x,\widehat{z}) + \varepsilon$$

for all k. We can find $\underline{z}_k$ such that

$$\Upsilon(\lambda;x_k,\widehat{z}) = G_{(\underline{z}_k)}(\lambda;x_k,\widehat{z}), \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda^-}\Upsilon(\lambda;x,\widehat{z}) = -\|\underline{z}_k-\widehat{z}\|^p.$$

If $\underline{z}_k$ is bounded, then up to a subsequence it converges to $\underline{z}_\infty$, and since G is upper semicontinuous,

$$\limsup_{k\to\infty}\Upsilon(\lambda;x_k,\widehat{z}) = \limsup_{k\to\infty} G_{(\underline{z}_k)}(\lambda;x_k,\widehat{z}) \le G_{(\underline{z}_\infty)}(\lambda;x,\widehat{z}) \le \Upsilon(\lambda;x,\widehat{z})$$

which is a contradiction. If $\underline{z}_k$ is unbounded, then up to a subsequence, for $\lambda'\in(\kappa,\lambda)$,

$$\begin{aligned}\Upsilon(\lambda';x_k,\widehat{z}) &\ge \Upsilon(\lambda;x_k,\widehat{z}) - (\lambda-\lambda')\frac{\mathrm{d}}{\mathrm{d}\lambda^-}\Upsilon(\lambda;x_k,\widehat{z})\\ &\ge \Upsilon(\lambda;x,\widehat{z}) + \varepsilon + (\lambda-\lambda')\|\underline{z}_k-\widehat{z}\|^p \to \infty\end{aligned}$$

as $k\to\infty$. Therefore

$$\begin{aligned}\lim_{k\to\infty} F_{(x_k)}(\lambda',\widehat{x}) &= \lim_{k\to\infty}\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;x_k,\widehat{Z}) \mid \widehat{X}=\widehat{x}\right] - \lambda'\|x_k-\widehat{x}\|^p\\ &= \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\lim_{k\to\infty}\Upsilon(\lambda;x_k,\widehat{Z}) \mid \widehat{X}=\widehat{x}\right] - \lambda'\|x-\widehat{x}\|^p = \infty.\end{aligned}$$

This contradicts with $\Phi(\lambda',\widehat{x})<\infty$.

We can thus construct $\overline{Z},\underline{Z}$ which depends on λ, $\widehat{Z}$ and x. Now we have

$$F_{(x)}(\lambda;\widehat{x}) = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Upsilon(\lambda;x,\widehat{Z}) \mid \widehat{X}=\widehat{x}\right] - \lambda\|x-\widehat{x}\|^p.$$

It is upper semicontinuous in x because each $\Upsilon(\lambda;x,\widehat{z})$ is upper semicontinuous in x, and the finite sum of upper semicontinuous functions is upper semicontinuous. Moreover,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^+}F_{(x)}(\lambda;\widehat{x}) = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\Upsilon(\lambda;x,\widehat{Z}) \mid \widehat{X}=\widehat{x}\right] - \|x-\widehat{x}\|^p = -\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\|\overline{Z}-\widehat{Z}\|^p \mid \widehat{X}=\widehat{x}\right] - \|x-\widehat{x}\|^p \to -\infty$$

as $x \to \infty$. By Lemma EC.4 (II) we can find $\overline{x}$ and $\underline{x}$ such that

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\Phi(\lambda;\widehat{x}) = -\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\|\overline{Z}-\widehat{Z}\|^p \mid \widehat{X}=\widehat{x}\right] - \|\overline{x}-\widehat{x}\|^p, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda^-}\Phi(\lambda;\widehat{x}) = -\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\|\underline{Z}-\widehat{Z}\|^p \mid \widehat{X}=\widehat{x}\right] - \|\underline{x}-\widehat{x}\|^p,$$

$$\Phi(\lambda;\widehat{x}) = F_{(\underline{x})}(\lambda;\widehat{x}) = F_{(\overline{x})}(\lambda;\widehat{x}).$$

By constructing these for every $\widehat{x}$ in the support of $\widehat{\mathbb{P}}_{\widehat{X}}$, we have $\overline{X}$, $\underline{X}$, $\overline{Z}$, $\underline{Z}$ such that $((\overline{X},\overline{Z}),(\widehat{X},\widehat{Z})) \sim \overline{\gamma}$, $((\underline{X},\underline{Z}),(\widehat{X},\widehat{Z})) \sim \underline{\gamma}$, where

$$\overline{\gamma} = \sum_{k=1}^{K}\sum_{i=1}^{n_k}\widehat{p}_{ki}\delta_{((\overline{x}_k,\overline{z}_{ki}),(\widehat{x}_k,\widehat{z}_{ki}))}, \qquad \underline{\gamma} = \sum_{k=1}^{K}\sum_{i=1}^{n_k}\widehat{p}_{ki}\delta_{((\underline{x}_k,\underline{z}_{ki}),(\widehat{x}_k,\widehat{z}_{ki}))}.$$

We use notations $\overline{\gamma}_1, \underline{\gamma}_1, \overline{\gamma}_2, \underline{\gamma}_2$ similar as in the proof of Theorem 1.

Now we have both

$$\begin{aligned}
h(\lambda) &= \lambda\rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\Phi(\lambda;\widehat{X})\right] \\
&= \lambda\rho^p + \mathbb{E}_{\overline{\gamma}_1}\left[F_{(\overline{X})}(\lambda;\widehat{X})\right] \\
&= \lambda\rho^p + \mathbb{E}_{\overline{\gamma}_1}\left[\mathbb{E}_{\overline{\gamma}_2}\left[\Upsilon(\lambda;\overline{X},\widehat{Z}) \mid (\overline{X},\widehat{X})\right] - \lambda\|\overline{X}-\widehat{X}\|^p\right] \\
&= \lambda\rho^p + \mathbb{E}_{\overline{\gamma}_1}\left[\mathbb{E}_{\overline{\gamma}_2}\left[G_{(\overline{Z})}(\lambda;\overline{X},\widehat{Z}) \mid (\overline{X},\widehat{X})\right] - \lambda\|\overline{X}-\widehat{X}\|^p\right] \\
&= \lambda\rho^p + \mathbb{E}_{\overline{\gamma}_1}\left[\mathbb{E}_{\overline{\gamma}_2}\left[\Psi(f(\overline{X}),\overline{Z}) - \lambda\|\overline{Z}-\widehat{Z}\|^p \mid (\overline{X},\widehat{X})\right] - \lambda\|\overline{X}-\widehat{X}\|^p\right] \\
&= \lambda\left(\rho^p - \overline{d}\right) + \mathbb{E}_{\overline{\mathbb{P}}}\left[\Psi(f(\overline{X}),\overline{Z})\right], \\
h(\lambda) &= \lambda\left(\rho^p - \underline{d}\right) + \mathbb{E}_{\underline{\mathbb{P}}}\left[\Psi(f(\underline{X}),\underline{Z})\right],
\end{aligned}$$

and

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\lambda^+}h(\lambda) &= \rho^p + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda^+}\Phi(\lambda;\widehat{X})\right] \\
&= \rho^p + \mathbb{E}_{\overline{\gamma}_1}\left[-\mathbb{E}_{\overline{\gamma}_2}\left[\|\overline{Z}-\widehat{Z}\|^p \mid (\overline{X},\widehat{X})\right] - \|\overline{X}-\widehat{X}\|^p\right] \\
&= \rho^p - \overline{d}, \\
\frac{\mathrm{d}}{\mathrm{d}\lambda^-}h(\lambda) &= \rho^p - \underline{d}.
\end{aligned}$$

At $\lambda = \lambda^*$, h is minimized, so $\frac{\mathrm{d}}{\mathrm{d}\lambda^-}h(\lambda^*) \le 0 \le \frac{\mathrm{d}}{\mathrm{d}\lambda^+}h(\lambda^*)$. Therefore there exists $q^* \in [0,1]$, such that

$$q^*\left(\rho^p - \overline{d}\right) + (1-q^*)\left(\rho^p - \underline{d}\right) = 0.$$

Then if we denote $\gamma^* = q^*\overline{\gamma} + (1-q^*)\underline{\gamma}$, then

$$\mathbb{E}_{((X,Z),(\widehat{X},\widehat{Z}))\sim\gamma^*}\left[\|X-\widehat{X}\|^p + \|Z-\widehat{Z}\|^p\right] = q^*\overline{d} + (1-q^*)\underline{d} = \rho^p.$$

Therefore, $\mathbb{P}^* = \gamma^*_{(X,Z)} = q^*\overline{\mathbb{P}} + (1-q^*)\underline{\mathbb{P}}$ is feasible, and

$$\mathbb{E}_{\mathbb{P}^*}[\Psi(f(X),Z)] = q^*\mathbb{E}_{\overline{\mathbb{P}}}[\Psi(f(\overline{X}),\overline{Z})] + (1-q^*)\mathbb{E}_{\overline{\mathbb{P}}}[\Psi(f(\overline{X}),\overline{Z})] = h(\lambda^*) = v^f_{\mathsf{D}} = v^f_{\mathsf{P}}$$

it is optimal.

Note that this optimal solution is

$$\mathbb{P}^* = \sum_{k=1}^{K}\sum_{i=1}^{n_k}\widehat{p}_{ki}\left(q^*\delta_{(\overline{x}_k,\overline{z}_{ki})} + (1-q^*)\delta_{(\underline{x}_k,\underline{z}_{ki})}\right).$$

Now we first consider the following linear optimization problem,

$$\begin{aligned}
\sup_{\{q_k\}_k\subset[0,1]} \quad & \mathbb{E}_{(X,Z)\sim\mathbb{P}}[\Psi(f(X),Z)] \\
& \text{where } \mathbb{P} = \sum_{k=1}^{K}\sum_{i=1}^{n_k}\widehat{p}_{ki}\left(q_k\delta_{(\overline{x}_k,\overline{z}_{ki})} + (1-q_k)\delta_{(\underline{x}_k,\underline{z}_{ki})}\right), \\
\text{s.t.} \quad & \mathbb{E}_{((X,Z),(\widehat{X},\widehat{Z}))\sim\gamma}\left[\|X-\widehat{X}\|^p + \|Z-\widehat{Z}\|^p\right] \le \rho^p \\
& \text{where } \gamma = \sum_{k=1}^{K}\sum_{i=1}^{n_k}\widehat{p}_{ki}\left(q_k\delta_{((\overline{x}_k,\overline{z}_{ki}),(\widehat{x}_k,\widehat{z}_{ki}))} + (1-q_k)\delta_{((\underline{x}_k,\underline{z}_{ki}),(\widehat{x}_k,\widehat{z}_{ki}))}\right).
\end{aligned}$$

The feasible domain is not empty because $q_k = q^*$ gives a feasible solution $\mathbb{P}^*$. The constraints and the target function are all linear functions of q_k, so the inf can be attained at the vertices of the feasible domain, and thus we can find k_0 such that $q_k = 1$ or 0 whenever $k \neq k_0$. So, we have found another optimal solution

$$\mathbb{P} = \sum_{k\neq k_0}\sum_{i=1}^{n_k}\widehat{p}_{ki}\delta_{(x^*_k,z^*_{ki})} + \sum_{i=1}^{n_{k_0}}\widehat{p}_{i_0 j}\left(q\delta_{(\overline{x}_{k_0},\overline{z}_{k_0 i})} + (1-q)\delta_{(\underline{x}_{k_0},\underline{z}_{k_0 i})}\right).$$

where $(x^*_k, z^*_{ki}) = (\overline{x}_k, \overline{z}_{ki})$ or $(\underline{x}_k, \underline{z}_{ki})$ depending only on k. Note that the marginal $\mathbb{P}_X$ is supported over at most $I+1$ points. □

EC.4. Proofs for Section 4

Proof of Corollary 1. Since $\Psi(\cdot, z)$ is affine for each z, Ψ can be written as

$$\Psi(w,z) = \ell^z(w), \qquad \ell^z(w) = \beta^{z\top}w + b^z.$$

Here ℓ^z is an affine function with gradient $\beta^z \in \mathcal{D}^*$ and intercept $b^z \in \mathbb{R}$. Then

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(w,\widehat{Z}) \mid \widehat{X} = \widehat{x}_k\right] = \frac{1}{\sum_{i=1}^{n_k}\widehat{p}_{ki}}\sum_{i=1}^{n_k}\widehat{p}_{ki}\Psi(w,\widehat{z}_{ki}) = \frac{1}{\widehat{p}_k}\sum_{i=1}^{n_k}\widehat{p}_{ki}\ell^{\widehat{z}_{ki}}(w)$$

Denote

$$\beta_k := \frac{1}{\widehat{p}_k}\sum_{i=1}^{n_k}\widehat{p}_{ki}\beta^{\widehat{z}_{ki}}, \qquad b_k := \frac{1}{\widehat{p}_k}\sum_{i=1}^{n_k}\widehat{p}_{ki}b^{\widehat{z}_{ki}},$$

and

$$\ell_k(w) := \frac{1}{\widehat{p}_k}\sum_{i=1}^{n_k}\widehat{p}_{ki}\ell^{\widehat{z}_{ki}}(w) = \beta_k^\top w + b_k, \tag{EC.12}$$

which is an affine function of w. Therefore, $\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(w,\widehat{Z}) \mid \widehat{X}=\widehat{x}_k\right] = \ell_k(w)$ is affine. We have

$$\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(f(x),\widehat{Z}) \mid \widehat{X}=\widehat{x}_k\right] - \lambda\|x-\widehat{x}_k\|^p\right\} = \sup_{x\in\mathcal{X}}\left\{\ell_k(f(x)) - \lambda\|x-\widehat{x}_k\|^p\right\}.$$

Suppose $f:\mathcal{X}\to\mathcal{D}$ is an affine decision rule, then $f(x) = B^\top x + \delta$, and

$$\ell_k(f(x)) - \ell_k(f(\widehat{x}_k)) = \beta_k^\top(f(x) - f(\widehat{x}_k)) = \beta_k^\top B^\top(x-\widehat{x}_k).$$

Thus, the supremum over x can be computed explicitly as

$$\begin{aligned}\sup_{x\in\mathcal{X}}\left\{\ell_k(f(x)) - \lambda\|x-\widehat{x}_k\|^p\right\} &= \ell_k(f(\widehat{x}_k)) + \sup_{x\in\mathcal{X}}\left\{(B\beta_k)^\top(x-\widehat{x}_k) - \lambda\|x-\widehat{x}_k\|^p\right\}\\ &= \ell_k(f(\widehat{x}_k)) + \sup_{t\geq 0}\left\{\|B\beta_k\|_* t - \lambda t^p\right\}.\end{aligned}$$

Define a convex function $R_p:\mathbb{R}_+^2\to\mathbb{R}\cup\{+\infty\}$ by

$$R_p(\lambda,\mu) := \sup_{t\geq 0}\left\{\mu t - \lambda t^p\right\} = \begin{cases}\infty\mathbf{1}\{\lambda<\mu\}, & p=1,\\ \lambda(p-1)\left(\frac{\mu}{\lambda p}\right)^{\frac{p}{p-1}}, & p>1.\end{cases} \tag{EC.13}$$

Then

$$\sup_{x\in\mathcal{X}}\left\{\ell_k(f(x)) - \lambda\|x-\widehat{x}_k\|^p\right\} = \ell_k(f(\widehat{x}_k)) + R_p\left(\lambda, |B\beta_k|\right),$$

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(f(x),\widehat{Z}) \mid \widehat{X}\right] - \lambda\|x-\widehat{X}\|^p\right\}\right] = \sum_{k=1}^K\left(\sum_{i=1}^{n_k}\widehat{p}_{ki}\right)\left[\ell_k(f(\widehat{x}_k)) + R_p\left(\lambda, \|B\beta_k\|_*\right)\right].$$

Note that R_p is a convex function in λ and B, $\ell_k(f(\widehat{x}_k)) = \ell_k(B^\top\widehat{x}_k + \delta)$ is affine in B and δ, so the right-hand side of the last expression is convex in λ and B as well. Hence (6) is a convex program:

$$\inf_{\lambda\geq 0,(B,\delta)\in\Theta}\left\{\lambda\rho^p + \sum_{k=1}^K\widehat{p}_k\left[\ell_k(B^\top\widehat{x}_k+\delta)) + R_p\left(\lambda, \|B\beta_k\|_*\right)\right]\right\},$$

where ℓ_k is an affine function defined by (EC.12). □

Proof of Corollary 2. We start with sup over z:

$$\sup_{z\in\mathcal{Z}}\left\{\Psi(w,z)-\lambda\|z-\widehat{z}_{ki}\|^2\right\}=\Psi(w,\widehat{z}_{ki})+\sup_{z\in\mathcal{Z}}\left\{(A^\top w+\alpha)^\top(z-\widehat{z}_{ki})-\lambda\|z-\widehat{z}_{ki}\|^2\right\}$$
$$=\Psi(w,\widehat{z}_{ki})+\sup_{\tilde{z}\in\mathcal{Z}}\left\{(A^\top w+\alpha)^\top\tilde{z}-\lambda\|\tilde{z}\|^2\right\}.$$

By the linearity of Ψ in z,

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\Psi(w,z)-\lambda\|z-\widehat{Z}\|^2\right\}\mid\widehat{X}=\widehat{x}_k\right]=\Psi(w,\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}[\widehat{Z}\mid\widehat{X}=\widehat{x}_k])+\sup_{\tilde{z}\in\mathcal{Z}}\left\{(A^\top w+\alpha)^\top\tilde{z}-\lambda\|\tilde{z}\|^2\right\}$$
$$=\Psi(w,\bar{z}_k)+\sup_{\tilde{z}\in\mathcal{Z}}\left\{(A^\top w+\alpha)^\top\tilde{z}-\lambda\|\tilde{z}\|^2\right\}$$
$$=\sup_{\tilde{z}\in\mathcal{Z}}\left\{\Psi(w,\bar{z}_k+\tilde{z})-\lambda\|\tilde{z}\|^2\right\}$$
$$=\sup_{z\in\mathcal{Z}}\left\{\Psi(w,z)-\lambda\|z-\bar{z}_k\|^2\right\}.$$

where we define $\bar{z}_k=\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}[\widehat{Z}\mid\widehat{X}=\widehat{x}_k]$. Next, we take the supremum in x with decision $w=f(x)=B^\top x+\delta$. Note that

$$\Psi(w,z)=\begin{pmatrix}w^\top & 1\end{pmatrix}\begin{pmatrix}A & \beta\\ \alpha^\top & b\end{pmatrix}\begin{pmatrix}z\\ 1\end{pmatrix}\qquad w=\begin{pmatrix}B^\top & \delta\end{pmatrix}\begin{pmatrix}x\\ 1\end{pmatrix}\quad\Longrightarrow\quad\Psi(f(x),z)=\begin{pmatrix}x^\top & 1\end{pmatrix}\begin{pmatrix}B & \mathbf{0}\\ \delta^\top & 1\end{pmatrix}\begin{pmatrix}A & \beta\\ \alpha^\top & b\end{pmatrix}\begin{pmatrix}z\\ 1\end{pmatrix}.$$

We thus express the supremum in x by

$$y_k:=\sup_{x\in\mathcal{X},z\in\mathcal{Z}}\Psi(w,z)-\lambda\|z-\bar{z}_k\|^2-\lambda\|x-\widehat{x}_k\|^2$$
$$=\sup_{x\in\mathcal{X},z\in\mathcal{Z}}\begin{pmatrix}x^\top & z^\top & 1\end{pmatrix}\begin{pmatrix}B & \mathbf{0}\\ O & \mathbf{0}\\ \delta^\top & 1\end{pmatrix}\begin{pmatrix}O & A & \beta\\ \mathbf{0}^\top & \alpha^\top & b\end{pmatrix}\begin{pmatrix}x\\ z\\ 1\end{pmatrix}-\lambda\begin{pmatrix}x^\top & z^\top & 1\end{pmatrix}\begin{pmatrix}I & O & -\widehat{x}_k\\ O & I & -\bar{z}_k\\ -\widehat{x}_k^\top & -\bar{z}_k^\top & \|\widehat{x}_k\|^2+\|\bar{z}_k\|^2\end{pmatrix}\begin{pmatrix}x\\ z\\ 1\end{pmatrix}.$$

We have transformed (D) into

$$\inf_{\substack{(B,\delta)\in\Theta\\ \lambda\geq 0,\{y_k\}_k\subset\mathbb{R}}}\quad\lambda\rho^2+\sum_{k=1}^K\widehat{p}_k y_k$$
$$\text{s.t.}\quad X_k\succeq O$$

where

$$X_k:=\lambda\begin{pmatrix}I & O & -\widehat{x}_k\\ O & I & -\bar{z}_k\\ -\widehat{x}_k^\top & -\bar{z}_k^\top & \|\bar{z}_k\|^2+\|\widehat{x}_k\|^2\end{pmatrix}+y_k\begin{pmatrix}O & O & \mathbf{0}\\ O & O & \mathbf{0}\\ \mathbf{0}^\top & \mathbf{0}^\top & 1\end{pmatrix}-\frac{1}{2}\left[\begin{pmatrix}B & \mathbf{0}\\ O & \mathbf{0}\\ \delta^\top & 1\end{pmatrix}\begin{pmatrix}O & A & \beta\\ \mathbf{0}^\top & \alpha^\top & b\end{pmatrix}+\begin{pmatrix}O & \mathbf{0}\\ A^\top & \alpha\\ \beta^\top & b\end{pmatrix}\begin{pmatrix}B^\top & O & \delta\\ \mathbf{0}^\top & \mathbf{0}^\top & 1\end{pmatrix}\right]$$

$$= \begin{pmatrix} \lambda I & -\frac{1}{2}BA & -\frac{1}{2}B\beta - \lambda\widehat{x}_k \\ -\frac{1}{2}(BA)^\top & \lambda I & -\frac{1}{2}(A^\top\delta+\alpha) - \lambda\bar{z}_k \\ -\frac{1}{2}(B\beta)^\top - \lambda\widehat{x}_k^\top & -\frac{1}{2}(A^\top\delta+\alpha)^\top - \lambda\bar{z}_k^\top & y_k - \beta^\top\delta - b + \lambda\|\bar{z}_k\|^2 + \lambda\|\widehat{x}_k\|^2 \end{pmatrix} \succeq O. \tag{EC.14}$$

Since X_k is affine in λ, y_k, B, δ, this is a semidefinite program. □

Proof of Corollary 3. We first study the dual formulation. For $\Psi(w,z) = \|w-z\|^2$, it holds that

$$\sup_{z\in\mathcal{Z}}\{\|w-z\|^2 - \lambda\|z-\widehat{z}\|^2\} = \begin{cases} \frac{\lambda}{\lambda-1}\|w-\widehat{z}\|^2 & \lambda > 1 \\ 0 & \lambda = 1 \text{ and } w = \widehat{z} \\ \infty & \text{otherwise.} \end{cases}$$

Restricting to $\lambda > 1$, the conditional expectation is

$$\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\{\|w-z\|^2 - \lambda\|z-\widehat{Z}\|^2\} \mid \widehat{X}\right] = \frac{\lambda}{\lambda-1}\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\|w-\widehat{Z}\|^2 \mid \widehat{X}\right] = \frac{\lambda}{\lambda-1}\left(\left\|w - \mathbb{E}[\widehat{Z}\mid\widehat{X}]\right\|^2 + \mathbb{V}\mathrm{ar}[\widehat{Z}\mid\widehat{X}]\right).$$

With affine decision rule, $w = f(x) = B^\top x + \delta$. So

$$\begin{aligned} &\sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\{\|B^\top x+\delta-z\|^2 - \lambda\|z-\widehat{Z}\|^2\} \mid \widehat{X}\right] - \lambda\|x-\widehat{X}\|^2\right\} \\ &= \sup_{x\in\mathcal{X}}\left\{\frac{\lambda}{\lambda-1}\left(\left\|B^\top x+\delta-\mathbb{E}[\widehat{Z}\mid\widehat{X}]\right\|^2 + \mathbb{V}\mathrm{ar}[\widehat{Z}\mid\widehat{X}]\right) - \lambda\|x-\widehat{X}\|^2\right\}. \end{aligned}$$

For each $\widehat{x}_k$, define y_k by

$$\begin{aligned} \lambda y_k &= \sup_{x\in\mathcal{X}}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\{\Psi(f(x),z) - \lambda\|z-\widehat{Z}\|^2\} \mid \widehat{X}=\widehat{x}_k\right] - \lambda\|x-\widehat{x}_k\|^2\right\} \\ &= \sup_{x\in\mathcal{X}}\left\{\frac{\lambda}{\lambda-1}\left(\left\|B^\top x+\delta-\mathbb{E}[\widehat{Z}\mid\widehat{X}=\widehat{x}_k]\right\|^2 + \mathbb{V}\mathrm{ar}[\widehat{Z}\mid\widehat{X}=\widehat{x}_k]\right) - \lambda\|x-\widehat{x}_k\|^2\right\}. \end{aligned}$$

Denote $\mu_k = \mathbb{E}[\widehat{Z}\mid\widehat{X}=\widehat{x}_k]$ and $\sigma_k = \mathbb{V}\mathrm{ar}[\widehat{Z}\mid\widehat{X}=\widehat{x}_k]$. The dual problem can thus be written as

$$\begin{aligned} \inf_{\substack{(B,\delta)\in\Theta \\ \lambda\geq 1 \\ \{y_k\}\in\mathbb{R}^K}} \quad & \lambda\rho^2 + \lambda\sum_k \widehat{p}_k y_k \\ \text{s.t.} \quad & (\lambda-1)y_k \geq \left\|B^\top x+\delta-\mu_k\right\|^2 + \sigma_k - (\lambda-1)\|x-\widehat{x}_k\|^2, \qquad \forall x\in\mathbb{R}^d, k\in[K]. \end{aligned}$$

The constraint can be written as a quadratic form of x, so it is equivalent to

$$\begin{pmatrix} (\lambda-1)I - BB^\top & -(\lambda-1)\widehat{x}_k + B(\mu_k-\delta) \\ -(\lambda-1)\widehat{x}_k^\top + (\mu_k-\delta)^\top B^\top & (\lambda-1)(y_k + \|\widehat{x}_k\|^2) - \|\delta-\mu_k\|^2 - \sigma_k \end{pmatrix} \succeq O \qquad k\in[K].$$

Using the standard lifting method, we rewrite it as

$$\begin{pmatrix} (\lambda-1)I - W & -(\lambda-1)\widehat{x}_k + B\mu_k - u \\ -(\lambda-1)\widehat{x}_k^\top + \mu_k^\top B^\top - u^\top & (\lambda-1)(y_k + \|\widehat{x}_k\|^2) - t + 2\delta^\top \mu_k - \|\mu_k\|^2 - \sigma_k \end{pmatrix} \succeq O \qquad \forall k \in [K],$$

$$\begin{pmatrix} W - BB^\top & u - B\delta \\ u^\top - (B\delta)^\top & t - \delta^\top \delta \end{pmatrix} \succeq O$$

for some $W \in \mathbb{R}^{d\times d}$, $u \in \mathbb{R}^d$, $t \in \mathbb{R}$. The latter constraint is equivalent to

$$\begin{pmatrix} W & B & u \\ B^\top & I & \delta \\ u^\top & \delta^\top & t \end{pmatrix} \succeq O,$$

hence $t \geq 0$. The proof is thus completed. □

Proof of Corollary 4. Using Theorem 1, we can write (9) in the dual form:

$$v_{\mathsf{D}} = \inf_{\lambda \geq 0, \mathbb{h} \in \mathbb{H}} \left\{ \lambda\rho + \mathbb{E}_{\widehat{\Phi}\sim\widehat{\mathbb{Q}}_\Phi} \left[\sup_{\phi \in \mathbb{H}^m} \left\{ \mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}} \left[\Psi(\langle \mathbb{h}, \phi\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi} \right] - \lambda \|\phi - \widehat{\Phi}\| \right\} \right] \right\}.$$

Here, we used the fact that the metric on $\mathcal{Z}$ is the infinity metric. Since $\Psi(\cdot, z)$ is Lipschitz and convex, so is $\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}[\Psi(\cdot, \widehat{Z}) \mid \widehat{\Phi} = \widehat{\phi}]$ for $\widehat{\mathbb{Q}}_{\widehat{\Phi}}$-a.e. $\widehat{\phi}$. Moreover, $\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}}$ is a bounded linear map, so $\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}[\Psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi} = \widehat{\phi}]$ is also a convex Lipschitz function for $\widehat{\mathbb{Q}}_{\widehat{\Phi}}$-a.e. $\widehat{\phi}$.. Therefore

$$\begin{aligned} &\sup_{\phi \in \mathbb{H}^m} \left\{ \mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}} \left[\Psi(\langle \mathbb{h}, \phi\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi} \right] - \lambda \|\phi - \widehat{\Phi}\| \right\} \\ &\qquad = \mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}} \left[\Psi(\langle \mathbb{h}, \widehat{\Phi}\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi} \right] + \infty \mathbf{1} \left\{ \lambda < \|\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}[\Psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi}]\|_{\mathrm{Lip}} \right\}. \end{aligned}$$

We now compute this Lipschitz norm. We claim for any convex Lipschitz function $\psi : \mathbb{R}^m \to \mathbb{R}$, it holds that

$$\|\psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}})\|_{\mathrm{Lip}(\mathbb{H}^m)} = \|\mathbb{h}\|_{\mathbb{H}} \|\psi\|_{\mathrm{Lip}(\mathbb{R}^m)}.$$

It is trivial to see that the equality holds when $\|\mathbb{h}\|_{\mathbb{H}} = 0$, and that the left-hand side is bounded by the right-hand side. To see that this bound is achievable, we first observe that $\langle \mathbb{h}, a\mathbb{h}\rangle = \|\mathbb{h}\|_{\mathbb{H}}^2 a$ for any $a \in \mathbb{R}^m$, where $a\mathbb{h} = \{a_i \mathbb{h}\}_{i\in[m]}$. We now take any $a, b \in \mathbb{R}^m$ and $a \neq b$, then

$$\begin{aligned} \frac{|\psi(a) - \psi(b)|}{\|a-b\|} = \frac{\left| \psi\left(\left\langle \mathbb{h}, \frac{a\mathbb{h}}{\|\mathbb{h}\|_{\mathbb{H}}^2} \right\rangle \right) - \psi\left(\left\langle \mathbb{h}, \frac{b\mathbb{h}}{\|\mathbb{h}\|_{\mathbb{H}}^2} \right\rangle \right) \right|}{\|a-b\|} &\leq \|\psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}})\|_{\mathrm{Lip}(\mathbb{H}^m)} \frac{\left\| \frac{a\mathbb{h}}{\|\mathbb{h}\|_{\mathbb{H}}^2} - \frac{b\mathbb{h}}{\|\mathbb{h}\|_{\mathbb{H}}^2} \right\|}{\|a-b\|} \\ &= \|\psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}})\|_{\mathrm{Lip}(\mathbb{H}^m)} \frac{1}{\|\mathbb{h}\|_{\mathbb{H}}}. \end{aligned}$$

Taking supremum over $x, y \in \mathbb{R}^m$ yields the desired equality. Combined, we can simplify v_{D} to

$$\begin{aligned}
v_{\mathsf{D}} &= \inf_{\lambda\geq 0, \mathbb{h}\in\mathbb{H}} \left\{\lambda\rho + \mathbb{E}_{\widehat{\Phi}\sim\widehat{\mathbb{Q}}_{\Phi}}\left[\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}\left[\Psi(\langle \mathbb{h}, \widehat{\Phi}\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi}\right] + \infty \mathbf{1}\left\{\lambda < \|\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}[\Psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi}]\|_{\mathrm{Lip}}\right\}\right]\right\} \\
&= \inf_{\mathbb{h}\in\mathbb{H}} \left\{\mathbb{E}_{\widehat{\Phi}\sim\widehat{\mathbb{Q}}_{\Phi}}\left[\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}\left[\Psi(\langle \mathbb{h}, \widehat{\Phi}\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi}\right]\right] + \rho \sup_{\widehat{\phi}\in\operatorname{supp}\widehat{\mathbb{Q}}_{\widehat{\Phi}}} \left\|\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}\left[\Psi(\langle \mathbb{h}, \cdot\rangle_{\mathbb{H}}, \widehat{Z}) \mid \widehat{\Phi} = \widehat{\phi}\right]\right\|_{\mathrm{Lip}}\right\} \\
&= \inf_{\mathbb{h}\in\mathbb{H}} \left\{\mathbb{E}_{(\widehat{\Phi},\widehat{Z})\sim\widehat{\mathbb{Q}}}\left[\Psi(\langle \mathbb{h}, \widehat{\Phi}\rangle_{\mathbb{H}}, \widehat{Z})\right] + \rho\|\mathbb{h}\|_{\mathbb{H}} \sup_{\widehat{\phi}\in\operatorname{supp}\widehat{\mathbb{Q}}_{\widehat{\Phi}}} \left\|\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{Q}}_{\widehat{Z}|\widehat{\Phi}}}\left[\Psi(\cdot, \widehat{Z}) \mid \widehat{\Phi} = \widehat{\phi}\right]\right\|_{\mathrm{Lip}}\right\} \\
&= \inf_{\mathbb{h}\in\mathbb{H}} \left\{\mathbb{E}_{(\widehat{X},\widehat{Z})\sim\widehat{\mathbb{P}}}\left[\Psi(\mathbb{h}(\widehat{X}), \widehat{Z})\right] + \rho\|\mathbb{h}\|_{\mathbb{H}} \sup_{\widehat{x}\in\operatorname{supp}\widehat{\mathbb{P}}_{\widehat{X}}} \left\|\mathbb{E}_{\widehat{Z}\sim\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\Psi(\cdot, \widehat{Z}) \mid \widehat{X} = \widehat{x}\right]\right\|_{\mathrm{Lip}}\right\}.
\end{aligned}$$

A representer theorem for vector-valued RKHS can be derived similarly to the scalar-valued version in (Shafieezadeh-Abadeh et al., 2019, theorem 27), thus the optimal $\mathbb{h}$ is given by

$$\mathbb{h}^{\star} = \sum_k \boldsymbol{K}(\cdot, \widehat{x}_k)\beta_k, \qquad \beta_k \in \mathbb{R}^m.$$

The proof is identical, so we omit the details. It is now a finite-dimensional convex optimization problem over $\{\beta_k\}_{k\in[K]}$. To derive (10), we compute

$$\begin{aligned}
\mathbb{h}(\widehat{x}) &= \sum_k \boldsymbol{K}(\widehat{x}, \widehat{x}_k)\beta_k, \\
\|\mathbb{h}\|_{\mathbb{H}}^2 &= \sum_{j,k\in[K]} \langle \boldsymbol{K}(\cdot, \widehat{x}_k)\beta_k, \boldsymbol{K}(\cdot, \widehat{x}_j)\beta_j\rangle = \sum_{j,k\in[K]} \beta_j^{\top}\boldsymbol{K}(\widehat{x}_j, \widehat{x}_k)\beta_k.
\end{aligned}$$

We thus conclude (10). □

Proof of Theorem 3. First, we show that $\cap_k I_k(x)$ is nonempty. To begin with, each $I_k(x)$ is nonempty, because the definition of ϕ_k implies

$$\varphi_k(w_k) \leq \phi_k \leq \lambda^*\|x - x_k\| + \phi_k,$$

so $w_k \in I_k(x)$. Note that each $I_k(x)$ is an interval since it is the sub-level set of a convex function φ_k. To prove they have a nonempty intersection, it suffices to show they pairwise intersect. For instance, we show here that $I_1(x)$ and $I_2(x)$ intersect by contradiction. Suppose I_1 and I_2 are disjoint. Since $w_1 \in I_1(x)$, $w_2 \in I_2(x)$, we know that I_1 and I_2 are disjoint if and only if we can find w_3 in between w_1 and w_2 outside both intervals. This implies that

$$\varphi_1(w_3) > \lambda^*\|x - x_3\| + \phi_1 \geq \lambda^*\|x - x_1\| + \varphi_1(w_1),$$

$$\varphi_1(w_3) > \lambda^*\|x - x_3\| + \phi_1 \geq \lambda^*\|x - x_1\| + \varphi_1(w_2) - \lambda^*\|x_1 - x_2\|,$$
$$\varphi_2(w_3) > \lambda^*\|x - x_3\| + \phi_2 \geq \lambda^*\|x - x_2\| + \varphi_2(w_2),$$
$$\varphi_2(w_3) > \lambda^*\|x - x_3\| + \phi_2 \geq \lambda^*\|x - x_2\| + \varphi_2(w_1) - \lambda^*\|x_1 - x_2\|.$$

Since w_3 is between w_1 and w_2, we can find $\alpha, \beta \in [0,1]$ with $\alpha + \beta = 1$ and $w_3 = \alpha w_1 + \beta w_2$. By multiplying the first/fourth inequality with α and the second/third inequality with β then taking the sum, we have

$$\begin{aligned}(\varphi_1 + \varphi_2)(w_3) &> \lambda^*(\|x - x_1\| + \|x - x_2\|) + \alpha(\varphi_1 + \varphi_2)(w_1) + \beta(\varphi_1 + \varphi_2)(w_2) - \lambda^*\|x_1 - x_2\| \\ &\geq \alpha(\varphi_1 + \varphi_2)(w_1) + \beta(\varphi_1 + \varphi_2)(w_2),\end{aligned}$$

using the triangle inequality. However, this contradicts the convexity of $\varphi_1 + \varphi_2$.

Next, we prove that any decision rule in the intersection $\cap_k I_k$ is optimal. For every $f \in \mathcal{F}$, let $\widehat{f} = f|_{\widehat{\mathcal{X}}} \in \widehat{\mathcal{F}}$ be the restriction of f on the set $\widehat{X}$, then

$$\begin{aligned}&\inf_{\lambda \geq 0}\left\{\lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x \in \mathcal{X}}\left\{\varphi(f(x); \lambda, \widehat{X}) - \lambda\|x - \widehat{X}\|\right\}\right]\right\} \\ &\geq \inf_{\lambda \geq 0}\left\{\lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{x \in \widehat{\mathcal{X}}}\left\{\varphi(f(x); \lambda, \widehat{X}) - \lambda\|x - \widehat{X}\|\right\}\right]\right\} \\ &= \inf_{\lambda \geq 0}\left\{\lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{1 \leq k \leq K}\left\{\varphi(\widehat{f}(x_k); \lambda, \widehat{X}) - \lambda\|x_k - \widehat{X}\|\right\}\right]\right\} \geq v_{\widehat{\mathsf{D}}}.\end{aligned} \tag{EC.15}$$

By taking the infimum over $f \in \mathcal{F}$, we would have $v_{\mathsf{D}} \geq v_{\widehat{\mathsf{D}}}$. On the other hand, for the minimizer λ^* and $\widehat{f}^* \in \widehat{\mathcal{F}}$ of (13), let $f \in \mathcal{F}$ be an extension in $\cap_k I_k(x)$, then for every x we have

$$\varphi_k(f(x)) - \lambda^*\|x - \widehat{x}\| \leq \max_{1 \leq k \leq K}\left\{\varphi(\widehat{f}(\widehat{x}_k); \lambda^*, \widehat{x}) - \lambda^*\|x_k - \widehat{x}\|\right\}.$$

Therefore,

$$\begin{aligned}&\lambda^*\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{1 \leq k \leq K}\left\{\varphi(\widehat{f}(x_k); \lambda^*, \widehat{X}) - \lambda^*\|x_k - \widehat{X}\|\right\}\right] \\ &\geq \lambda^*\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x \in \mathcal{X}}\left\{\varphi(f(x); \lambda^*, \widehat{X}) - \lambda^*\|x - \widehat{X}\|\right\}\right] \geq v_{\mathsf{D}}.\end{aligned}$$

Thus $v_{\mathsf{D}} = v_{\widehat{\mathsf{D}}}$.

Finally, we show the necessity of the interval condition. Suppose $f^* \in \mathcal{F}$ is an optimal policy to the problem (11) with optimal dual value λ^*. By (EC.15), λ^* and the restriction $\widehat{f}^* = f|_{\widehat{\mathcal{X}}} \in \widehat{\mathcal{F}}$ are

also an optimal dual value and an optimal policy to the problem (13). To show that $f^*(x) \in \cap_k I_k(x)$, we prove by contradiction. Suppose for some $x \in \mathcal{X}$ and some $k \in [K]$, $f^*(x) \notin I_k(x)$. This means

$$\varphi(f^*(x);\lambda^*,\widehat{x}_k) = \varphi_k(f^*(x)) > \lambda^*\|x-\widehat{x}_k\| + \phi_k = \lambda^*\|x-\widehat{x}_k\| + \max_j \left\{\varphi_k(w_j) - \lambda^*\|\widehat{x}_k - \widehat{x}_j\|\right\}.$$

That is, there exists $k \in [K]$ such that for all $j \in [K]$,

$$\varphi(f^*(x);\lambda^*,\widehat{x}_k) - \lambda^*\|x-\widehat{x}_k\| > \varphi(f^*(\widehat{x}_j);\lambda^*,\widehat{x}_k) - \lambda^*\|\widehat{x}_k - \widehat{x}_j\|.$$

Then

$$\begin{aligned} v_{\mathsf{D}} &= \lambda^*\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\sup_{x\in\mathcal{X}}\left\{\varphi(f^*(x);\lambda^*,\widehat{X}) - \lambda^*\|x-\widehat{X}\|\right\}\right] \\ &> \lambda^*\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{j\in[K]}\left\{\varphi(f^*(\widehat{x}_j);\lambda^*,\widehat{X}) - \lambda^*\|\widehat{x}_j-\widehat{X}\|\right\}\right] \geq v_{\widehat{\mathsf{D}}}, \end{aligned}$$

which contradicts with $v_{\mathsf{D}} = v_{\widehat{\mathsf{D}}}$. Therefore, we must have $f^*(x) \in \cap_k I_k(x)$ for all $x \in \mathcal{X}$, which completes the proof of the theorem. □

Proof of Remark 2. Recall that we consider the linear loss function $\Psi(w,z) = w^\top z$, where the decision space $\mathcal{D} = B_1$ is the unit disk in $\mathbb{R}^2$, and $\mathcal{X} = \mathcal{Z} = \mathbb{R}^2$. Suppose $\widehat{\mathbb{P}}$ satisfies

$$\widehat{\mathbb{P}}_{\widehat{X}} = \frac{1}{3}\sum_{k=1}^{3}\delta_{\widehat{x}_k}, \qquad \widehat{x}_k = \left(\cos\frac{2k\pi}{3}, \sin\frac{2k\pi}{3}\right), \qquad \mathbb{E}\left[\widehat{Z} \mid \widehat{X}\right] = \frac{2\sqrt{3}}{3}\widehat{X}.$$

$\widehat{X} = \{\widehat{x}_1, \widehat{x}_2, \widehat{x}_3\}$ consists of three points on the unit circle that form an equilateral triangle (see Figure 5). Let us first solve the in-sample dual problem (13). Observe that

$$\sup_{z\in\mathcal{Z}}\{\Psi(w,z) - \lambda\|z-\widehat{z}\|\} = w^\top\widehat{z} + \infty\mathbf{1}\{\|w\| > \lambda\},$$

so we can find

$$\begin{aligned} \varphi(w;\lambda,\widehat{x}_j) = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{\Psi(w,z) - \lambda\|z-\widehat{Z}\|\right\} \mid \widehat{X} = \widehat{x}_j\right] &= w^\top\mathbb{E}\left[\widehat{Z} \mid \widehat{X} = \widehat{x}_j\right] + \infty\mathbf{1}\{\|w\| > \lambda\} \\ &= \frac{2\sqrt{3}}{3}w^\top\widehat{x}_j + \infty\mathbf{1}\{\|w\| > \lambda\}. \end{aligned}$$

(13) becomes

$$\min_{\substack{\widehat{f}\in\widehat{\mathcal{F}} \\ \lambda\geq\|\widehat{f}(\widehat{x}_k)\|}}\left\{\lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{1\leq k\leq 3}\left\{\frac{2\sqrt{3}}{3}\widehat{f}(\widehat{x}_k)^\top\widehat{X} - \lambda\|\widehat{x}_k - \widehat{X}\|\right\}\right]\right\}$$

$$
\begin{aligned}
&= \min_{\substack{\{w_k\}_k \in \mathcal{D} \\ \lambda \geq \|w_k\|}} \left\{ \lambda\rho + \frac{1}{3}\sum_{j=1}^{3} \max_{1\leq k\leq 3} \left\{ \frac{2\sqrt{3}}{3} w_k^\top \widehat{x}_j - \lambda \|\widehat{x}_k - \widehat{x}_j\| \right\} \right\} \\
&\geq \min_{\substack{\{w_k\}_k \in \mathcal{D} \\ \lambda \geq \|w_k\|}} \left\{ \lambda\rho + \frac{1}{3}\sum_{j=1}^{3} \frac{2\sqrt{3}}{3} w_k^\top \widehat{x}_j \right\} \\
&\geq \min_{\lambda\geq 0} \left\{ \lambda\rho - \frac{2\sqrt{3}}{3}(\lambda \wedge 1) \right\}.
\end{aligned}
$$

The last inequality used $\|w_k\| \leq \lambda \wedge 1$ and $\|\widehat{x}_j\| = 1$. Provided $0 < \rho < \frac{2\sqrt{3}}{3}$, the above is minimized at $\lambda = 1$, and we obtain a lower bound $\rho - \frac{2\sqrt{3}}{3}$ for (13). Note that $\|\widehat{x}_k - \widehat{x}_j\| = \sqrt{3}\mathbf{1}\{j \neq k\}$, and $\widehat{x}_k^\top \widehat{x}_j = \mathbf{1}\{j = k\} - \frac{1}{2}\mathbf{1}\{j \neq k\}$. One can directly check that this minimal value is achieved by $w_k = -\widehat{x}_k$ and $\lambda^* = 1$, thus (13) is solved.

Now we show that the extension in Theorem 3 does not exist. By the definition of $I_k(x)$, we know $w \in I_k(x)$ iff

$$
\varphi_k(w) - \lambda^* \|x - \widehat{x}_k\| \leq \phi_k = \max_j \{\varphi_k(w_j) - \lambda^* \|\widehat{x}_k - \widehat{x}_j\|\}.
$$

With $\lambda^* = 1$ and $w_j = -\widehat{x}_j$, we rewrite the above as

$$
\frac{2\sqrt{3}}{3} w^\top \widehat{x}_k - \|x - \widehat{x}_k\| \leq \max_j \left\{ \frac{2\sqrt{3}}{3} w_j^\top \widehat{x}_k - \|\widehat{x}_k - \widehat{x}_j\| \right\} = \max_j \left\{ -\frac{2\sqrt{3}}{3} \widehat{x}_j^\top \widehat{x}_k - \|\widehat{x}_k - \widehat{x}_j\| \right\} = -\frac{2\sqrt{3}}{3}.
$$

When $x = 0$, $\|x - \widehat{x}_k\| = 1$, so

$$
I_k(0) = \left\{ w \in \mathcal{D} : \frac{2\sqrt{3}}{3} w^\top \widehat{x}_k - 1 \leq -\frac{2\sqrt{3}}{3} \right\} = \left\{ w \in \mathcal{D} : w^\top \widehat{x}_k \leq \frac{\sqrt{3}}{2} - 1 \right\}.
$$

Since $\frac{\sqrt{3}}{2} - 1 < 0$, no such w exists in $\cap_k I_k(0)$, because $\widehat{x}_1, \widehat{x}_2, \widehat{x}_3$ form an equilateral triangle. One can also see from Figure 5 that the intersection of $I_k(x)$ is empty. □

EC.5. Proofs for Examples in Section 4

Proof of Example 5. Since f is real-valued and Ψ is convex in w, we use Theorem 3, so it has the following reformulation

$$
\inf_{\substack{\widehat{f}:\widehat{\mathcal{X}}\to\mathbb{R} \\ \lambda\geq 0}} \left\{ \lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}} \left[\max_{1\leq k\leq K} \left\{ \varphi(\widehat{f}(\widehat{x}_k); \lambda, \widehat{X}) - \lambda \|\widehat{x}_k - \widehat{X}\| \right\} \right] \right\}
$$

with

$$
\varphi(w; \lambda; \widehat{x}) = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}} \left[\sup_{z\in\mathcal{Z}} \left\{ |w - z| - \lambda \|z - \widehat{Z}\| \right\} \mid \widehat{X} = \widehat{x} \right].
$$

For any $\lambda < 1$, the supremum over z is infinite, hence $\varphi(w;\lambda,\widehat{x}) = \infty$. For $\lambda \geq 1$, the supremum is attained at $z = \widehat{Z}$, so

$$\varphi(w;\lambda;\widehat{x}) = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[|w - \widehat{Z}| \,\middle|\, \widehat{X} = \widehat{x}\right] + \infty \mathbf{1}\{\lambda < 1\}.$$

Thus, we reach the following reformulation,

$$\inf_{\substack{\widehat{f}:\widehat{\mathcal{X}}\to\mathbb{R} \\ \lambda\geq 1}} \left\{\lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{1\leq k\leq K}\left\{\mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[|\widehat{f}(\widehat{x}_k) - \widehat{Z}| \,\middle|\, \widehat{X} = \widehat{x}\right] - \lambda\|\widehat{x}_k - \widehat{X}\|\right\}\right]\right\}.$$

This can be transformed into a linear programming problem

$$\begin{aligned}
\inf_{\substack{\{w_k\}_k,\{y_k\}_k\subset\mathbb{R} \\ \{c_{kji}\}_{kji}\subset\mathbb{R},\lambda\geq 1}} \quad & \lambda\rho + \sum_{k=1}^{K} y_k \\
\text{s.t.} \quad & y_j \geq \sum_{i=1}^{n_j} \widehat{p}_{ki}(c_{kji} - \lambda\|\widehat{x}_k - \widehat{x}_j\|) && \forall j,k \in [K], \\
& c_{kji} \geq w_k - \widehat{z}_{ji} && \forall k,j \in [K], i \in [n_j], \\
& c_{kji} \geq \widehat{z}_{ji} - w_k && \forall k,j \in [K], i \in [n_j].
\end{aligned}$$

□

Proof of Example 6. Recall that the problem could be reformulated as

$$\inf_{\substack{\widehat{f}:\widehat{\mathcal{X}}\to\mathbb{R} \\ \lambda\geq 0}} \left\{\lambda\rho + \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{X}}}\left[\max_{1\leq k\leq K}\left\{\varphi(\widehat{f}(\widehat{x}_k);\lambda,\widehat{X}) - \lambda\|\widehat{x}_k - \widehat{X}\|\right\}\right]\right\},$$

where

$$\varphi(w;\lambda;\widehat{x}) = \mathbb{E}_{\widehat{\mathbb{P}}_{\widehat{Z}|\widehat{X}}}\left[\sup_{z\in\mathcal{Z}}\left\{-wz^\top\begin{pmatrix} w \\ 1\end{pmatrix} - \lambda\|z - \widehat{Z}\|\right\} \mid \widehat{X} = \widehat{x}\right].$$

When $\mathcal{Z}$ is equipped with the usual ℓ^p norm $\|\cdot\|_{\mathcal{Z}}$, the supremum over z in the definition of φ is infinite if $\|w\begin{pmatrix} w & 1\end{pmatrix}\|_* > \lambda$, where $\|\cdot\|_*$ is the dual norm of $\|\cdot\|_{\mathcal{Z}}$, otherwise the supremum is achieved at $z = \widehat{Z}$. Therefore

$$\varphi(w;\lambda;\widehat{x}_k) = -\begin{pmatrix} w^2 & w\end{pmatrix}\bar{z}_k + \infty\mathbf{1}\{\|\begin{pmatrix} w^2 & w\end{pmatrix}\|_* > \lambda\}.$$

Hence, we obtain the reformulation (14). Recall that the first component of z represents the price sensitivity coefficient, which is negative.

When $p = 1$, this can be written as the following quadratic constraint program:

$$\begin{aligned}
\inf_{\{w_k\}_k, \lambda \geq 0} \quad & \lambda\rho + \sum_{j\in[K]} \widehat{p}_j c_j \\
\text{s.t.} \quad & c_j + \left(w_k^2 \;\; w_k\right) \bar{z}_k + \lambda \|\widehat{x}_k - \widehat{x}_j\| \geq 0 && \forall j, k \in [K], \\
& w_k \leq \lambda && \forall k \in [K], \\
& w_k^2 \leq \lambda && \forall k \in [K].
\end{aligned}$$

When $p = \infty$, this can also be written as a quadratic constraint program:

$$\begin{aligned}
\inf_{\{w_k\}_k, \lambda \geq 0} \quad & \lambda\rho + \sum_{j\in[K]} \widehat{p}_j c_j \\
\text{s.t.} \quad & c_j + \left(w_k^2 \;\; w_k\right) \bar{z}_k + \lambda \|\widehat{x}_k - \widehat{x}_j\| \geq 0 && \forall j, k \in [K], \\
& w_k^2 + w_k \leq \lambda && \forall k \in [K].
\end{aligned}$$

When $p = 2$, this is written as

$$\begin{aligned}
\inf_{\{w_k\}_k, \lambda \geq 0} \quad & \lambda\rho + \sum_{j\in[K]} \widehat{p}_j c_j \\
\text{s.t.} \quad & c_j + \left(w_k^2 \;\; w_k\right) \bar{z}_k + \lambda \|\widehat{x}_k - \widehat{x}_j\| \geq 0 && \forall j, k \in [K], \\
& w_k^4 + w_k^2 \leq \lambda^2 && \forall k \in [K].
\end{aligned}$$

By introducing auxiliary variable $y_k = w_k^2$, this can be represented as a second-order conic programming:

$$\begin{aligned}
\inf_{\{w_k\}_k, \{y_k\}_k, \lambda \geq 0} \quad & \lambda\rho + \sum_{j\in[K]} \widehat{p}_j c_j \\
\text{s.t.} \quad & c_j + \left(w_k^2 \;\; w_k\right) \bar{z}_k + \lambda \|\widehat{x}_k - \widehat{x}_j\| \geq 0 && \forall j, k \in [K], \\
& y_k \geq w_k^2 && \forall k \in [K], \\
& y_k^2 + w_k^2 \leq \lambda^2 && \forall k \in [K].
\end{aligned}$$

□

Proof of Example 7. (D) and (4) are reduced to

$$\inf_{\substack{(B,\delta)\in\Theta \\ \lambda\geq 0, \{y_k\}_k \subset \mathbb{R}}} \quad \lambda\rho^2 + \sum_{k=1}^{K} \widehat{p}_k y_k$$

$$\text{s.t.} \quad \begin{pmatrix} x^\top & z^\top & 1 \end{pmatrix} X_k \begin{pmatrix} x \\ z \\ 1 \end{pmatrix} \geq 0, \qquad \forall k \in [K], x \in \mathcal{X}, z \in \mathcal{Z}$$

$$\begin{pmatrix} C_\ell^\top B^\top & C_\ell^\top \delta - c_\ell \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} \leq 0, \qquad \forall \ell \in [L], x \in \mathcal{X}.$$

where X_k is a symmetric matrix defined in (EC.14) with $A = I$, $\alpha = \beta = \mathbf{0}$, and $b = 0$:

$$X_k = \begin{pmatrix} \lambda I & -\frac{1}{2}B & -\lambda \widehat{x}_k \\ -\frac{1}{2}B^\top & \lambda I & -\frac{1}{2}\delta - \lambda \bar{z}_k \\ -\lambda \widehat{x}_k^\top & -\frac{1}{2}\delta^\top - \lambda \bar{z}_k^\top & y_k + \lambda \|\bar{z}_k\|^2 + \lambda \|\widehat{x}_k\|^2 \end{pmatrix}.$$

By the S-lemma Yakubovich (1977); Pólik and Terlaky (2007), two set of constraints are equivalent to

$$\inf_{\substack{B \in \mathbb{R}^{d \times m}, \delta \in \mathbb{R} \\ \lambda \geq 0, \{y_k\}_k \subset \mathbb{R} \\ \{\mu_k\}_k, \{\nu_\ell\}_\ell \subset \mathbb{R}_+}} \quad \lambda \rho^2 + \sum_{k=1}^K \widehat{p}_k y_k$$

$$\text{s.t.} \quad \begin{pmatrix} x^\top & z^\top & 1 \end{pmatrix} X_k \begin{pmatrix} x \\ z \\ 1 \end{pmatrix} + \mu_k ((x - x_0)^\top \Sigma (x - x_0) - R) \geq 0, \qquad \forall k \in [K], x \in \mathbb{R}^d, z \in \mathcal{Z}$$

$$-\begin{pmatrix} C_\ell^\top B^\top & C_\ell^\top \delta - c_\ell \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} + \nu_\ell ((x - x_0)^\top \Sigma (x - x_0) - R) \geq 0, \quad \forall \ell \in [L], x \in \mathbb{R}^d.$$

Constraints can be written as the semidefinite form:

$$X_k + \mu_k \begin{pmatrix} \Sigma & O & -\Sigma x_0 \\ O & O & \mathbf{0} \\ -x_0^\top \Sigma & \mathbf{0}^\top & x_0^\top \Sigma x_0 - R \end{pmatrix} \succeq O, \qquad -\begin{pmatrix} O & \frac{1}{2} B C_\ell \\ \frac{1}{2} C_\ell^\top B^\top & C_\ell^\top \delta - c_\ell \end{pmatrix} + \nu_\ell \begin{pmatrix} \Sigma & -\Sigma x_0 \\ -x_0^\top \Sigma & x_0^\top \Sigma x_0 - R \end{pmatrix} \succeq O.$$

We thus completed the proof of this example. □

We also formulate Example 7 with $\mathfrak{M}$ being the Wasserstein ball. Together with strong duality results for Wasserstein distributionally robust optimization (Gao and Kleywegt, 2023; Zhang et al., 2025), (4) can be written as

$$\inf_{\substack{(B,\delta) \in \Theta \\ \lambda \geq 0, \{y_{ki}\}_{ki} \subset \mathbb{R}}} \quad \lambda \rho^2 + \sum_{k=1}^K \sum_{i=1}^{n_k} \widehat{p}_{ki} y_{ki}$$

$$\text{s.t.} \quad \begin{pmatrix} x^\top & z^\top & 1 \end{pmatrix} X_{ki} \begin{pmatrix} x \\ z \\ 1 \end{pmatrix} \geq 0, \qquad \forall k \in [K], i \in [n_k], x \in \mathcal{X}, z \in \mathcal{Z}$$

$$\begin{pmatrix} C_\ell^\top B^\top & C_\ell^\top \delta - c_\ell \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} \leq 0, \qquad \forall \ell \in [L], x \in \mathcal{X}.$$

where X_k is a symmetric matrix defined in (EC.14) with $A = I$, $\alpha = \beta = \mathbf{0}$, and $b = 0$:

$$X_{ki} = \begin{pmatrix} \lambda I & -\frac{1}{2}B & -\lambda \widehat{x}_k \\ -\frac{1}{2}B^\top & \lambda I & -\frac{1}{2}\delta - \lambda \widehat{z}_{ki} \\ -\lambda \widehat{x}_k^\top & -\frac{1}{2}\delta^\top - \lambda \widehat{z}_{ki}^\top & y_{ki} + \lambda \|\widehat{z}_{ki}\|^2 + \lambda \|\widehat{x}_k\|^2 \end{pmatrix}.$$

Again by the S-lemma (Yakubovich, 1977; Pólik and Terlaky, 2007), two set of constraints are equivalent to

$$\inf_{\substack{B \in \mathbb{R}^{d \times m}, \delta \in \mathbb{R} \\ \lambda \geq 0, \{y_{ki}\}_{ki} \subset \mathbb{R} \\ \{\mu_{ki}\}_{ki}, \{\nu_\ell\}_\ell \subset \mathbb{R}_+}} \quad \lambda \rho^2 + \sum_{k=1}^{K} \sum_{i=1}^{n_k} \widehat{p}_{ki} y_{ki}$$

$$\text{s.t.} \quad \begin{pmatrix} x^\top & z^\top & 1 \end{pmatrix} X_{ki} \begin{pmatrix} x \\ z \\ 1 \end{pmatrix} + \mu_{ki}((x - x_0)^\top \Sigma (x - x_0) - R) \geq 0, \quad \forall k \in [K], i \in [n_k], x \in \mathbb{R}^d, z \in \mathcal{Z}$$

$$-\begin{pmatrix} C_\ell^\top B^\top & C_\ell^\top \delta - c_\ell \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} + \nu_\ell((x - x_0)^\top \Sigma (x - x_0) - R) \geq 0, \qquad \forall \ell \in [L], x \in \mathbb{R}^d.$$

Constraints can be written in the semidefinite form:

$$X_{ki} + \mu_{ki} \begin{pmatrix} \Sigma & O & -\Sigma x_0 \\ O & O & \mathbf{0} \\ -x_0^\top \Sigma & \mathbf{0}^\top & x_0^\top \Sigma x_0 - R \end{pmatrix} \succeq O, \qquad -\begin{pmatrix} O & \frac{1}{2}BC_\ell \\ \frac{1}{2}C_\ell^\top B^\top & C_\ell^\top \delta - c_\ell \end{pmatrix} + \nu_\ell \begin{pmatrix} \Sigma & -\Sigma x_0 \\ -x_0^\top \Sigma & x_0^\top \Sigma x_0 - R \end{pmatrix} \succeq O.$$

This is again a positive semidefinite program.

EC.6. Additional Numerical Experiments

EC.6.1. Comparison of Shapley and Non-Shapley Extension

Next, in our second set of experiments, we aim to compare the performance among different extensions of the in-sample optimal policy that are optimal to the DRO with causal transport

distance, as discussed in Theorem 3 and Remark 3. We first compare the performance between $\mathsf{C}_{\mathsf{Lip}}$ and $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$. Figure EC.1 shows the relative differences in out-of-sample expected cost between $\mathsf{C}_{\mathsf{Lip}}$—a positive number indicates that $\mathsf{C}_{\mathsf{Lip}}$ outperforms $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$.

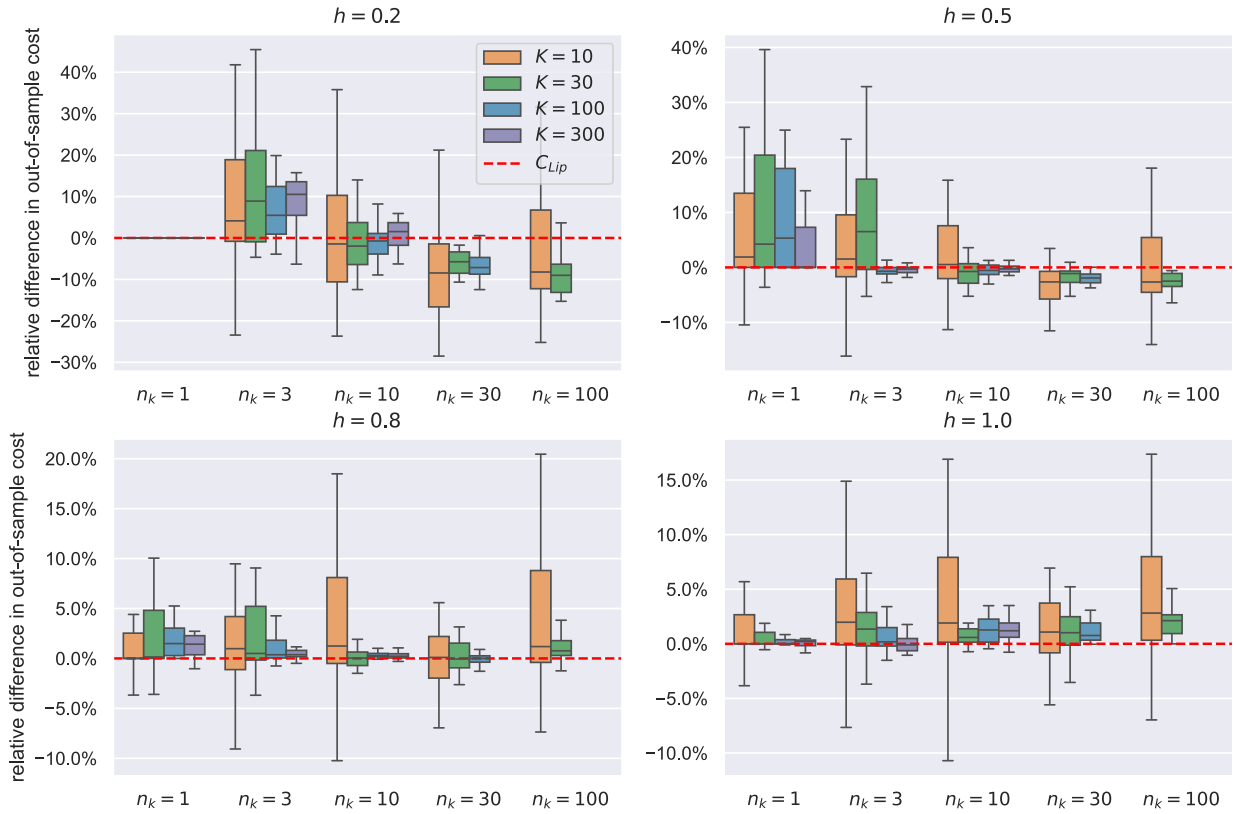

Figure EC.1 Boxplots of the relative differences in the out-of-sample performance between $\mathsf{C}_{\mathsf{Lip}}$ (baseline) and $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$

We observe that both $\mathsf{C}_{\mathsf{Lip}}$ and $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$ have their own competitive advantages. Specifically, $\mathsf{C}_{\mathsf{Lip}}$ demonstrates superior performance compared to $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$ when dealing with a relatively small sample size K, given the same n/K. On the other hand, for a fixed sample size K, $\mathsf{C}_{\mathsf{Lip}}$ outperforms $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$ when the ratio of n/K is low. This can be attributed to $\mathsf{C}_{\mathsf{Lip}}$ minimizing the ∞-norm, which leads to a more conservative approach than that of $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$. $\mathsf{C}_{\mathsf{Lip}}$ is more adept at managing situations with sparse data per covariate group. It minimizes the impact of potential outliers or extreme scenarios, which is helpful when individual covariate groups have fewer observations. However, this conservatism may reduce its effectiveness when the sample size is large.

EC.6.2. Comparison of Truncated and Non-truncated Policy

Moreover, we want to investigate further the relationship between different policies versus their truncated versions on the optimal region, as identified in Theorem 3. To begin with, we compare the performance of C_{TV} with $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$. Figure EC.2 shows the mean of the differences of out-of-sample costs between $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$ and C_{TV} with the same training and testing data set. A negative number implies that C_{TV} is outperformed by $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$. In general, $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$ has an advantage over C_{TV}, especially when the overage cost h is small or n/K is small. Their differences are not very large in general, as C_{TV} lies within the optimal region under most covariate values.

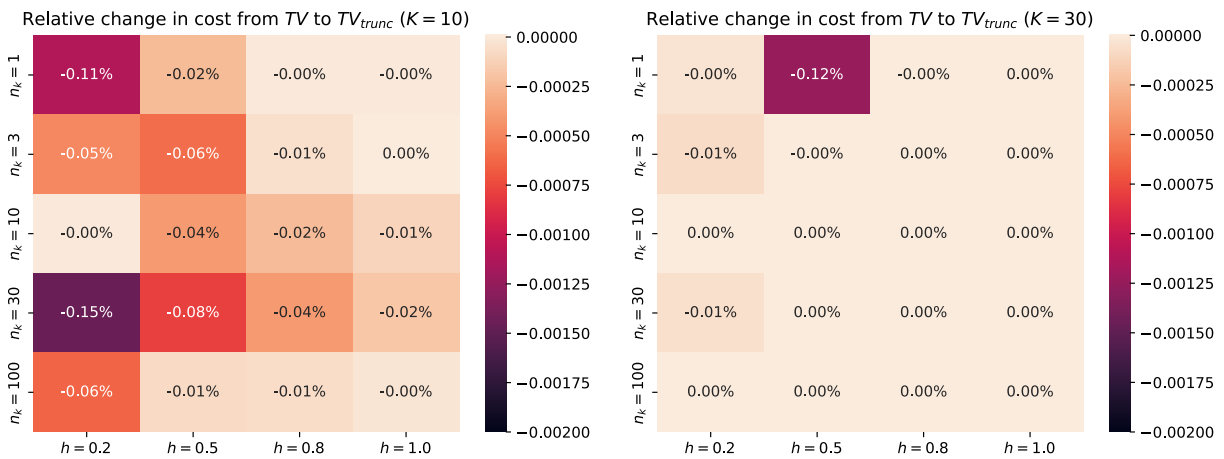

Figure EC.2 The differences in the out-of-sample performance between C_{TV} and $\mathsf{C}_{\mathsf{TV\text{-}trunc}}$

To deepen our understanding of the performance differences between policies and their truncated counterparts, we also compare the performance of empirical risk minimization using affine policy with ℓ^1 and ℓ^2 regularization ($\mathsf{ERM2}\,(\ell^1/\ell^2)$) in Ban and Rudin (2019) and their truncated versions. The results are shown in Figure EC.3. Setting $\mathsf{C}_{\mathsf{Lip}}$ as the baseline, the enhanced performance of the truncated versions emphasizes the efficacy of the optimal region.

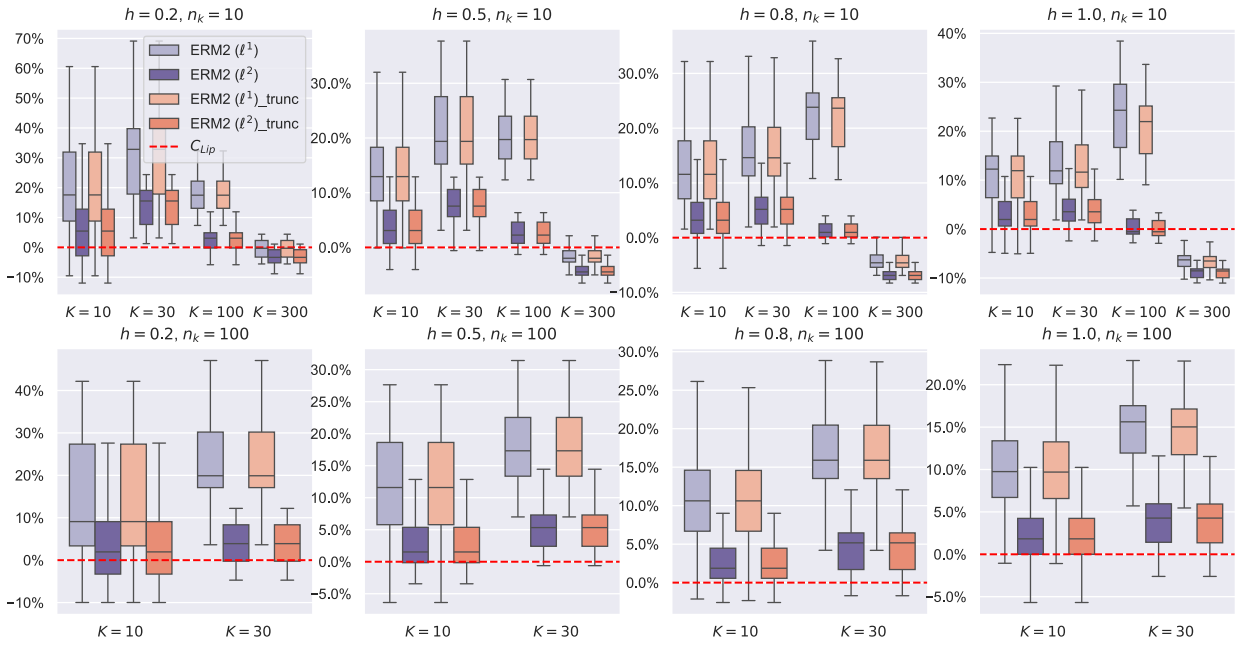

Figure EC.3 Boxplots of the relative differences in the out-of-sample performance between $\mathsf{C}_{\mathsf{Lip}}$ (baseline) and $\mathsf{ERM2}\,(\ell^1/\ell^2)$

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